Causal and Non-Causal Approaches to Treatment Effect Heterogeneity in Conjoint Experiments

An Application to Post-Conflict Interethnic Trust

Master Thesis

presented by

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Contents

1 Introduction **3**

2 Background **5**
- 2.1 Setup and Notation 5
- 2.2 Effect Dependency 8
- 2.3 Causal Effect Modification 12
- 2.4 Conjoint Experiments 15

3 Method **18**
- 3.1 Estimand 18
- 3.2 Estimation With Linear Models & Estimation Challenges 21
- 3.3 Estimation Strategy 23

4 Case Study **29**
- 4.1 Background & Theory 30
- 4.2 Data 31
- 4.3 Methodology 34
- 4.4 Results & Discussion 38

5 Concluding Remarks **40**

6 References **41**

A Appendix: Graphical Perspectives on Effect Dependency **46**

B Appendix: Simulation Study **47**
- B.1 Setup 48
- B.2 Models 50
- B.3 Results 51

C Appendix: Propensity Score Estimation **53**

How can researchers identify causal treatment effect modification when the effect modifier is not randomized? Researchers often study heterogeneous treatment effects by comparing how the treatment operates across subgroups defined by pre-treatment characteristics. Because these characteristics are rarely randomized, observed differences in treatment effects may reflect covariate imbalance rather than genuine causal effect modification. This issue is particularly salient in research using conjoint experiments, where the profiles respondents evaluate are randomized but the respondent characteristics used to define subgroups of interest are not. This thesis addresses how to estimate causal treatment effect modification when the effect modifier of interest is observational. I propose using inverse probability weighting as a form of nonparametric preprocessing: a flexible model estimates propensity scores for the effect modifier, and a simple parametric model is then estimated on the weighted data. Weighting balances covariates across modifier groups, allowing differences in treatment effects to be interpreted as causal effect modification rather than only as observed heterogeneity. This approach combines flexible propensity score estimation with interpretable outcome modeling and reduces model dependence, and reliance on restrictive functional form assumptions. I illustrate its implementation using original data from a conjoint experiment on post-conflict interethnic trust in Bosnia and Herzegovina, examining whether exposure to conflict-related sexual violence causally modifies the effect of ethnic group membership on interpersonal trust.

1 Introduction

Many social science questions concern not only whether a policy or intervention, commonly referred to as "treatment", has an effect, but also whether that effect depends on some third variable, an "effect modifier": a policy might work differently depending on the institutional context, or a program might be more effective for individuals with different background characteristics. Researchers in political science frequently investigate such heterogeneous effects in experimental and quasi-experimental designs. Studies using survey experiments and conjoint designs often examine whether treatment effects vary across respondents with different prior attitudes or socioeconomic characteristics (e.g. Bansak et al., 2016; Christensen et al., 2021; Franchino & Zucchini, 2014; Hainmueller & Hopkins, 2014; Hankinson, 2018; Leeper et al., 2020). Similarly, field and natural experiments frequently evaluate whether policy interventions have different impacts across social or institutional contexts (e.g. Enos et al., 2019; Hainmueller et al., 2017). In most of these studies, the treatment itself is randomized, but the characteristics used to define subgroups are not. This creates an important inferential challenge: distinguishing genuine causal effect modification from heterogeneity that reflects preexisting differences between units.

Conjoint experiments (Hainmueller et al., 2014) are a particularly prominent example of this challenge: while the profiles respondents evaluate are fully randomized, the respondent characteristics used to define subgroups, such as prior attitudes, identity, or experience are not, meaning that subgroup comparisons in conjoint studies, which are of considerable interest to researchers (cf. Bechtel et al., 2017; Hankinson, 2018; Leeper et al., 2020; Mummolo, 2016; Oliveros & Schuster, 2018), face the same identification problem as any other observational comparison.

A large body of literature studies heterogenous treatment effects (HTE), which broadly divide into data-driven approaches that search for HTE using regularization or machine learning (ML) techniques (e.g. Imai & Ratkovic, 2013; Künzel et al., 2019; Nie & Wager, 2021; Wager & Athey, 2018), and theory-driven approaches that build on domain knowledge to model interactions between treatments and prespecified covariates (e.g. Bansak, 2021; Christensen et al., 2021; Dong, 2015; Lee & Lee, 2025),[1] which ties into a rich literature on types of effect dependence (e.g. VanderWeele, 2009, 2015; VanderWeele & Robins, 2007; Vansteelandt et al., 2008). This thesis is situated within the second, theory-driven approach, and focuses on the identification problem that arises when the variables defining subgroups of interest are observational rather than randomized.

While variation in treatment effects is often substantively important and of interest to applied researchers, interpreting the nature of treatment effect heterogeneity is tricky. When the treatment, as well as the effect modifier, are randomly assigned, interpretation as causal interaction is straightforward. However, in most real-world empirical settings, potential effect modifiers are not randomly distributed. Researchers finding evidence of heterogeneity can thus only claim that the treatment is differently impactful, or works differently, for different segments of their sample, but cannot claim whether that difference is attributable to the effect modifier itself. Units self-select into different social categories, institutional contexts, or behavioral patterns, and this selection is often related to factors that also shape how they respond to the treatment. As a result, observed differences in treatment effects across groups may reflect preexisting differences rather than genuine causal effect modification. Apparent interactions between treatments and effect modifiers may arise because background characteristics influence both selection into the effect modifier and responsiveness to the treatment.

Standard analytical approaches, which typically rely on linear regression models with interaction terms, are ill-equipped to address this problem when either selection processes or treatment effect heterogeneity are complex or nonlinear. Models are also commonly poorly specified to make causal claims about effect modification, even under linearity. The motivating problem of this thesis is therefore how to recover causal treatment effect modification in settings where the effect modifier of interest was not randomized, and where observed effect heterogeneity may reflect covariate imbalance rather than true causal interaction. I propose using inverse probability weighting as "nonparametric preprocessing" (cf. Ho et al., 2007), where a flexible model that can dynamically model complex relationship, such as nonlinearity and interactions, is used to estimate propensity scores, and a simple parametric model is then estimated on the weighted data, so that weighting adjusts for preexisting covariate imbalance. This approach combines flexible propensity score estimation and interpretable outcome modelling, so that results remain easily interpretable, while being less model-dependent and less likely to violate functional form assumptions.

While the proposed approach applies to experimental designs generally, conjoint experiments are a natural and increasingly common setting where this problem arises, and the thesis gives particular attention to this design, showing how they are a special case of the general randomized controlled trial, and demonstrating the implementation of the proposed approach using original data from a conjoint experiment.

The remainder of the thesis proceeds as follows. In chapter 2, I begin by laying out the conceptual

[1] Hu (2023) provides an overview of both of these "camps".

background, distinguishing causal from non-causal effect modification and situating the problem within the broader literature on effect dependence, as well as introduce conjoint experiments and how they relate to the problem at hand. In chapter 3, I present the proposed estimation strategy based on inverse probability weighting with flexible propensity score estimation. I then illustrate the approach in chapter 4 using a conjoint experiment on wartime victimization and post-conflict interethnic trust conducted in Bosnia and Herzegovina.

2 Background

I study the problem of identifying and estimating causal treatment effect modification when the treatment is randomized but the potential effect modifier is not. By "causal treatment effect modification" I mean that the causal effect of the treatment on the outcome differs systematically across levels of a third variable, the effect modifier, such that intervention on the effect modifier would alter the treatment effect. While randomized experiments permit unbiased estimation of average treatment effects, assessing whether a non-randomized effect modifier varies the treatment effect causally is complicated by selection. In particular, background covariates may simultaneously influence selection into the effect modifier and treatment effect heterogeneity, inducing spurious interactions between treatment and effect modifier status.

In this section, I will start by laying out the problem from first principles, and situating this thesis within the broader literature on effect dependency. I will also disentangle causal effect modification from other kinds of effect heterogeneity, and consider the special case of conjoint experiments.

2.1 Setup and Notation

2.1.1 Potential Outcomes

Assume we have data on N individuals, sampled from some population, indexed as $i = 1, \dots, N$, and we want to learn the effect of some binary treatment T on some outcome Y. In the potential outcomes framework (Imbens & Rubin, 2015; Rubin, 1974, 2005), we imagine that each individual has two potential outcomes: $Y_i(1)$, the outcome we would observe if the individual were exposed to the treatment, and $Y_i(0)$, the outcome we would see if it they were instead left untreated (the "control" condition). Say subject i has a headache, and i decides to take a pill against headache. A while later, the headache is gone. Now, did the headache go away *because* i took the pill? The outcome of interest is i's observed health or headache status Y_i, and the treatment is whether i takes the pill ($T_i = 1$), or not ($T_i = 0$). To know whether the pill caused i's recovery, we would need to compare two states of the world: one in which i took the pill, i's outcome $Y_i(1)$, and one in which i did not; i's outcome $Y_i(0)$. If i takes the pill, only the first is observable; the second is a counterfactual. These "potential outcomes" capture the idea that treatment effects are defined at the individual level as the contrast

$$\tau_i = Y_i(1) - Y_i(0)$$

called the individual treatment effect (ITE). This quantity denotes the *actual* effect of taking the headache pill. However, in practice we only ever get to observe one of these outcomes, the realized outcome. This limitation is known as the "fundamental problem of causal inference" (Holland, 1986, p. 947):

$$Y_i = Y_i(T_i) = \begin{cases} Y_i(0) & \text{if } T_i = 0 \\ Y_i(1) & \text{if } T_i = 1 \end{cases}$$

This means we only ever get to observe what actually happens: if i takes the pill, then $T_i = 1$ and we observe $Y_i(1)$; the counterfactual $Y_i(0)$, what would have happened if i had not taken it, remains unobserved, and vice versa. Since that makes the ITE τ_i fundamentally unidentifiable, as it requires observing two mutually exclusive outcomes simultaneously, attention shifts to population-level quantities. A key quantity of interest here is the average treatment effect (ATE):

$$\bar{\tau} = \mathbb{E}[Y(1) - Y(0)]$$

Here, $\bar{\tau}$ denotes our expectation of the difference in outcomes between taking the pill, and not taking it. To identify the ATE, we rely on three assumptions. First, the Stable Unit Treatment Value Assumption (SUTVA) (e.g. Imbens & Rubin, 2015), which requires that there is a single, well-defined version of the treatment and that one individual's treatment status does not affect another's potential outcomes, meaning whether subject i takes the pill or not does not change which outcomes are available to another subject j. Second, if we randomize treatment assignment, then treatment is independent of the potential outcomes, $(Y_i(1), Y_i(0)) \perp\!\!\!\perp T_i$, since each individual's probability of treatment is unrelated to how they would respond to it. This is known as the independence or ignorability assumption. Third, randomization also ensures positivity, i.e. $0 < \Pr(T_i = 1) < 1 \; \forall \; i$, guaranteeing that each individual could in principle be observed under either condition. Together, these assumptions ensure that the simple difference-in-means estimator

$$\mathbb{E}[Y \mid T = 1] - \mathbb{E}[Y \mid T = 0]$$

is consistent and unbiased (Holland, 1986; Imai et al., 2010; Rubin, 1974). In our example, doing an experiment where we randomly assign subjects to the pill or a placebo identifies the ATE: the difference in average outcomes between groups then provides a consistent estimator of the causal effect of the pill. The reason this works is not the estimator itself, but the design. This makes a broader point: the potential outcomes framework shows that causal identification always rests on assumptions about the world that data alone cannot verify.

2.1.2 Directed Acyclic Graphs

Potential outcomes tell us how to define and estimate causal effects, but they do not tell us how variables relate to one another in the data-generating process. Directed acyclic graphs (DAGs, Pearl, 2000) fill this gap by providing a visual language for representing the researcher's assumptions about causal structures.

A DAG $G = (\mathbf{V}, \mathbf{E})$ is made up of a set of nodes (also called "vertices", or in this context "variables") $\mathbf{V}$, and $\mathbf{E}$, a set of directed edges $X \to Y$ where $X, Y \in \mathbf{V}$. While a simple directed graph may include cycles (e.g. $X \to Y$, $Y \to X$) denoting feedback loops or mutual causation, a DAG does not (hence *acyclic*). Each node $V \in \mathbf{V}$ represents a random variable. A directed edge $X \to Y$ indicates that X is a direct cause of Y. Note that the absence of an arrow between two nodes denotes the assumption that there is no such relationship between two nodes, which is an equally strong assumption. An example of a DAG is shown in 1.

$$\begin{tikzcd}
W \arrow[r] & Z \arrow[r] \arrow[d] & X \arrow[dl] \\
 & Y &
\end{tikzcd}$$

Figure 1: A directed acyclic graph (DAG)

We use kinship terminology to denote relationships inside the graph. For example, we might say that X and Z are both parents of Y, and that X and Y are both children of Z in figure 1. There are some common patterns in a DAG that are of substantive interest to causal inference: a *confounder* is a common cause (a "fork"). A *mediator* lies on a directed path, forming a chain. Lastly, a *collider* is a common effect of two or more other variables. A visual representation of each can be found in figure 2. A path is open, or "unblocked", at non-colliders, meaning confounders and mediators: in figure 2a & b, the path from T to Y is open through Z, T and Y are said to be *d-connected*. Paths are closed at colliders, so in 2c, T and Y are *d-separated*. Open paths may induce statistical associations between variables.

$$\begin{tikzcd}
 & Z \arrow[dl] \arrow[dr] & \\
T & & Y
\end{tikzcd}$$

(a) Z is a confounder.

$$\begin{tikzcd}
 & Z \arrow[dr] & \\
T \arrow[ur] & & Y
\end{tikzcd}$$

(b) Z is a mediator.

$$\begin{tikzcd}
 & Z & \\
T \arrow[ur] & & Y \arrow[ul]
\end{tikzcd}$$

(c) Z is a collider.

Figure 2: Common structures in a DAG.

Conditioning on a variable, i.e. introducing it into the analysis e.g. via regression control or stratification, has different effects depending on the variables role. Conditioning on a confounder blocks the spurious backdoor path and is generally desirable. Conditioning on a mediator blocks part of the causal effect itself, which may or may not be intended depending on the research question. Conditioning on a collider, by contrast, is hazardous: it opens a path that was previously closed, introducing a spurious association between variables that were otherwise unrelated. To summarize, conditioning on a non-collider blocks paths, conditioning on a collider opens them.

Confounding especially is a key concept, perhaps best understood through an example. Consider the contact hypothesis (Allport, 1954): the idea that intergroup contact reduces prejudice. Say we hypothesize the relationship looks like in figure 3a. We collect some data, looking at whether

people with more ethnic intergroup contact have lower prejudice, and we indeed observe a positive relationship: people who have more frequent outgroup contact exhibit lower prejudice toward members of the outgroup. However, in reality, the mechanism might look more like 3b, where prior attitudes toward the outgroup confound the relationship: subjects who have less prejudice to begin with, perhaps because of their social background or education, are more likely to have (or even actively seek) contact, and exhibit lower levels of prejudice. Failure to condition on this confounder biases our estimate of the true effect of outgroup contact; prior attitudes might wholly or partly account for the observed effect if left unaddressed.

(a) Outgroup contact reduces prejudice; the contact hypothesis.

(b) However, the relationship might be confounded.

Figure 3: A simplified DAG for the contact hypothesis, showing how failure to condition on confounders in observational data introduces bias.

Together, the potential outcomes framework and DAGs provide the conceptual foundation for the concepts and analyses that follow. The former defines what we mean by a causal effect, and the latter makes explicit the assumptions under which it can be identified. Both will prove indispensable when we turn to the question of whether, and for whom, causal effects differ.

2.2 Effect Dependency

Many social science questions concern not just whether a single treatment or intervention has an effect, but also whether the effects of multiple factors depend on one another. For example, a policy might be more effective in certain economic or political contexts, or a training program might work better for individuals with different background characteristics. Understanding these effect dependencies can help researchers better understand underlying mechanisms, and also to design more targeted interventions. Conceptually, there is now a third variable S (which we assume is binary for now), which is also assumed to be a cause of the outcome (cf. figure 4).[2] Since both T and S have a direct effect on the outcome Y, their respective effects might depend on the value of the other variable, or they might have a joint effect that is separate from their distinct effects.

In this section, I will attempt to disentangle the different kinds of effect dependency, effect modification (or "heterogeneity"), which is usually not considered as a causal quantity, and interaction, which does have a causal interpretation.

[2]Standard DAGs are agnostic to effect dependency, meaning that S is taken to be an effect modifier of $T \rightarrow Y$ because it also has a direct effect on Y. There are attempts to make effect dependency more explicit in DAGs, which I discuss in the appendix.

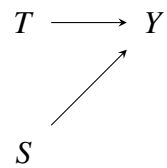


Figure 4: S is another cause of Y, and might thus vary $T \to Y$.

2.2.1 Effect Modification

Effect modification means that the effect of a treatment varies across levels of a third variable (VanderWeele, 2009). Formally,

$$\mathbb{E}[Y(1) - Y(0) \mid S = 1] \neq \mathbb{E}[Y(1) - Y(0) \mid S = 0],$$

where

$$\tau(s) = \mathbb{E}[Y(1) - Y(0) \mid S = s] \tag{1}$$

is called the "conditional average treatment effect", or CATE (e.g. Abrevaya et al., 2015). Effect modification by itself does not imply causation: it only requires the effect of the treatment T on the outcome Y to be identified, and then estimates the treatment effect separately for subgroups of interest defined by the effect modifier S. If $T \to Y$ is identified, we can estimate $\tau(s)$ as

$$\mathbb{E}[Y \mid T = 1, S = s] - \mathbb{E}[Y \mid T = 0, S = s].$$

As effect modification is not inherently a causal quantity, we can only claim heterogeneity: we can state that we observe a certain CATE at $S = s$, but we cannot claim that this is due to s: since we only estimate over the observed distribution, and its separate effect is not identified, we can not make a claim that an intervention that sets $S = s$ would causally alter the treatment effect.

2.2.2 Interaction

An "interaction" between two variables means that they have a joint effect that is distinct from their two effects considered separately (VanderWeele, 2009). Interaction is a causal quantity, and requires the two separate effect of T and S to be identified.[3] In the literature, the structure of the potential outcomes is commonly changed from $Y(t)$ to $Y(t, s)$ when discussing interaction: Previously, we considered our outcome to be merely a function of the treatment, but now we consider it to be a function of both T and S, $Y_i(T_i, S_i)$, and as such consider S as a secondary treatment to intervene on (VanderWeele, 2015). Instead of two, each unit i now has four potential outcomes, again only one of which is observed:

[3] Note that "interaction" is also often used in a purely statistical sense, where it is synonymous with effect dependency (cf. e.g. Christensen et al., 2021). In the context of this thesis, I use the word "interaction" in the causal sense defined by VanderWeele (2009), although it is not ideal that the terminology is so heavily conflated.

$$Y_i = Y_i(T_i, S_i) = \begin{cases} 1,1 & \text{if } T_i = 1,\ S_i = 1 \\ 1,0 & \text{if } T_i = 1,\ S_i = 0 \\ 0,1 & \text{if } T_i = 0,\ S_i = 1 \\ 0,0 & \text{if } T_i = 0,\ S_i = 0 \end{cases}$$

Formally, we consider two interventions T and S to interact if [4]

$$\mathbb{E}[Y(1,1) - Y(0,1)] \neq \mathbb{E}[Y(1,0) - Y(0,0)].$$

A question we might ask is: does the effect of T and S applied simultaneously differ from when they are applied individually (Keele & Stevenson, 2021)? We can formalize the average interaction effect (AIE, Egami & Imai, 2018) as

$$\mathbb{E}[Y(1,1)] - \mathbb{E}[Y(0,1)] - \mathbb{E}[Y(1,0)] + \mathbb{E}[Y(0,0)]. \tag{2}$$

Assuming $T \to Y$ and $S \to Y$ are both identified, we can estimate

$$\tau(t,s) = \mathbb{E}[Y \mid T = t, S = s],$$

which allows us to estimate the interaction effect as

$$\mathbb{E}[Y \mid T = 1, S = 1] - \mathbb{E}[Y \mid T = 0, S = 1] - \mathbb{E}[Y \mid T = 1, S = 0] + \mathbb{E}[Y \mid T = 0, S = 0] \tag{3}$$

We can rewrite this in a more intuitive way as a "difference-in-differences"

$$\begin{aligned} &\mathbb{E}\Big[\mathbb{E}[Y \mid T = 1, S = 1] - \mathbb{E}[Y \mid T = 0, S = 1]\Big] \\ -&\mathbb{E}\Big[\mathbb{E}[Y \mid T = 1, S = 0] - \mathbb{E}[Y \mid T = 0, S = 0]\Big] \end{aligned} \tag{4}$$

which we can read as "the difference in the effect of T when $S = 1$, and when $S = 0$". It is important to note that interaction is symmetrical, and we could also write this difference-in-differences the other way around as the difference in $S \to Y$ across T, which is simply another reformulation of eq. 3:

[4] Interaction is also scale dependent: whether two causes are said to interact depends on the scale on which the outcome is measured. For example, two variables may interact on the additive scale but not on the multiplicative scale, or vice versa. In this thesis I focus on interaction on the additive scale, which corresponds to the interaction effect (eq. 2), which connects naturally to the linear model (as I show further below), and is commonly used in the causal inference literature (e.g. VanderWeele, 2015).

$$\mathbb{E}\Big[\mathbb{E}[Y \mid T = 1, S = 1] - \mathbb{E}[Y \mid T = 1, S = 0]\Big] - \mathbb{E}\Big[\mathbb{E}[Y \mid T = 0, S = 1] - \mathbb{E}[Y \mid T = 0, S = 0]\Big].$$

Effect dependency (interaction or effect modification) is commonly assessed by fitting a regression model that includes a product term between the two exposures (Vansteelandt et al., 2008). Following VanderWeele (2009), we might think of effect modification in this context as

$$\mathbb{E}[Y(t) \mid S = s] = \alpha_0 + \alpha_1 T + \alpha_2 S + \alpha_3 (T \times S)$$

and of interaction as

$$\mathbb{E}[Y(t, s)] = \beta_0 + \beta_1 T + \beta_2 S + \beta_3 (T \times S).$$

The two models are really structurally the same, the only difference is in the estimated quantity and the interpretation of the coefficients: for effect modification, we simply estimate over the observed distribution of S, while for (causal) interaction we treat S as a second intervention and require its effect on Y to be identified as well.

To make the distinction more intuitive, here is an example.[5] Consider a voter mobilization experiment with two independently randomized treatments: T_1, whether a household receives a knock on the door from a canvasser encouraging them to vote, and T_2, whether they receive a mailer informing them that the majority of their neighbors voted in the last election. The outcome Y denotes whether a subject actually turned out to vote. Because both T_1 and T_2 are randomized, we can ask not only about the effect of each separately, but whether they interact; that is, whether receiving both produces a larger or smaller effect on turnout than the sum of the two individual effects would suggest. Perhaps the social norm mailer is more effective among those also contacted by a canvasser, because the visit raises the salience of the election. Due to randomization, any such heterogeneity reflects a genuine causal interaction, not a compositional artifact of who happens to receive which treatment. Past turnout S is a natural effect modifier for the canvasser treatment. People who did not vote in the last election may be more responsive to a canvasser visit than those who did, since habitual voters are likely to turn out regardless. We might therefore observe larger treatment effects among previous non-voters, but this difference is not necessarily caused by past turnout itself. Previous non-voters differ from habitual voters in many ways: age, socioeconomic status, civic engagement, etc. (which we can denote together as background factors $\mathbf{X}$) and it is these underlying differences, rather than past turnout per se, that likely drive the heterogeneity. Past turnout is thus an effect modifier: a variable across whose levels the treatment effect differs, without being the cause of that difference. Figure 5 shows the mechanism graphically.

[5] Inspired by papers on get-out-the-vote (GOTV) experiments such as Cardy (2005) or Murray and Matland (2014).

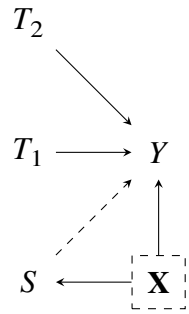


Figure 5: T_1 and T_2 interact, while S is an effect modifier. Note that the existence of $S \rightarrow Y$ is not of interest here because no causal claim is made about S, we simply observe that people who voted in the last election react differently to the treatment than those who did not. Whether or not we believe there is a $S \rightarrow Y$ is related to the distinction between causal and non-causal, or surrogate, effect modification, which I discuss in the following section.

2.3 Causal Effect Modification

But what if researchers observe effect modification by some observational third variable S, which they *would like* to make causal claims about, despite S being observational? Perhaps the term "effect modification" is somewhat misleading, as "modification" implies some sort of active intervention. It might be a good idea to reason about *effect modification* as defined above as *effect heterogeneity* instead. If we observe that the effect of T on Y differs across values of S, we can claim that the effect of T on Y varies across values of S, but we cannot claim whether that difference is attributable to S itself. This is because there might exist some other factors $\mathbf{X}$ that affect S, and also vary the treatment effect, so that if they are unaccounted for, variation in treatment effects due to $\mathbf{X}$ are attributed to S.

Switching back to single-treatment potential outcomes $Y_i = Y_i(T_i)$ since S is some covariate & not intervened on in this context, a simple way to formalize & estimate treatment effect heterogeneity is via CATEs (cf. eq. 1), where we only consider T as a treatment to invervene on and simply estimate over the observed distribution of S. We could then estimate CATE at different values of S, and take the difference as an estimate of treatment effect heterogeneity across values of S (or in classical terminology "effect modification by S"):

$$\Delta_{\text{CATE}}(S) = \tau(1) - \tau(0). \tag{5}$$

Again, here we can only be sure that the effect of T on Y differs across values of S, not whether that difference is attributable to S itself. However, CATEs as estimate of heterogeneity are often already of interest to applied researchers, and there is a wealth of data-driven approaches concerned with discovering and estimating treatment effect heterogeneity, as alluded to in the introduction (cf. e.g. Imai & Ratkovic, 2013; Künzel et al., 2019; Nie & Wager, 2021; Wager & Athey, 2018).

In applied research, however, researchers are also often interested in whether observed heterogeneity is attributable to S itself. Consider the following example: imagine we are researchers evaluating

a new economic policy and have the chance to run an experiment on a random sample of towns from some country. We randomly assign some towns to implement the policy and others to serve as controls, and we find an overall positive treatment effect: towns adopting the policy end up better off. We then hypothesize that the policy might work especially well in places with higher state capacity, measured e.g. as the number of public employees per capita, because they can implement it more effectively. When we examine the data, we do see that towns with more public employees appear to experience larger treatment effects. But does that mean we can recommend hiring additional public officials alongside implementing the policy to make it more effective? While the treatment is randomized, the number of public officials a town has is not. This means there might exist factors that both affect the number of public officials, and also vary the efficacy of the treatment, such as how rich a town was in the first place, confounding the observed heterogeneity. We can claim that our policy is differently impactful dependent on the number of public officials, but we cannot claim that difference is attributable to the number of public officials itself: figure 6 shows how this reduces to a plain confounding problem: we cannot claim causality, or "interaction", since the separate effect of state capacity is not identified, as it is confounded.

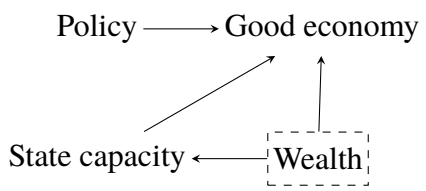


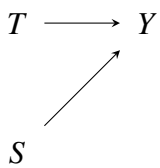
Figure 6: The separate effect of state capacity is not identified, since we do not account for confounders, like wealth. We can thus not make a causal claim about the way that it varies the effectiveness of our policy.

Conceptually, we can claim that the effect of T on Y differs across values of S, but we cannot claim whether that difference is attributable to S itself. This is due to the possible existence of background factors $\mathbf{X}$, which might affect selection into values of S, and also modify the treatment effect. This means that if $\mathbf{X}$ goes unaccounted for, difference in treatment effects due to $\mathbf{X}$ are attributed to S instead.

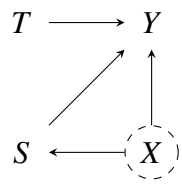
Graphically, the distinction becomes a little more clear. If we had randomized S, we would be dealing with simple interaction: the separate effects of T and S on Y are identified, and so is thus their joint effect, as can be seen in figure 7a. We can speak of S as a *causal effect modifier* of T's effect on Y. Consider next the settings from 7b & c: in the first, S is also a causal effect modifier, but its effect on Y is confounded by $\mathbf{X}$. In the latter, things are even more messy: $\mathbf{X}$ is a causal effect modifier, but S is not; however, $\mathbf{X}$ induces a statistical relationship that makes it look like S modifies the effect of T on Y. S becomes a *surrogate effect modifier*.[6]

From the perspective of VanderWeele (2009), in order to make a claim about whether the difference is attributable to S itself we would need the independence assumption to hold for both S and T, so that we can identify the (symmetric) causal interaction effect. However, in many settings, randomizing potential effect modifiers is not feasible, for example due to ethical or design constraints, or

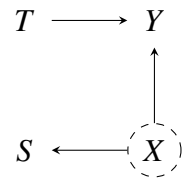
[6]The distinction between *causal* and *surrogate* effect modifier is coined by Hernán and Robins (2020). While VanderWeele and Robins (2007) have already clarified the different types of effect modification, the terminology of Hernán and Robins (2020) is in my opinion much clearer about the roles of the involved variables.



(a) Interaction (causal effect modification)



(b) Causal effect modification with confounding.



(c) Surrogate effect modification.

Figure 7: Types of effect modification. The dashed line indicates that X is unaccounted for.

heterogeneity might be discovered post-hoc, and researchers might wish to assess causality based on the observational data they have already collected instead of rerunning potentially costly experiments.

We can define causal treatment effect modification[7] as follows: the effect of a treatment T on some outcome Y differs across a third variable S, such that intervention on S would alter the effect of T on Y. This is effectively the same as causal interaction, the difference comes from the fact that S was not intervened on, and $S \to Y$ is not identified without additional steps and assumptions. Figure 8 shows intuitively how to classify types of effect dependency when T is random and $T \to Y$ is thus identified. It also highlights the point that the difference between causal and surrogate effect modification is mostly about what the researcher assumes about $S \to Y$: is there actually a separate effect of S on Y that is just not properly identified, due to S being observational, or is S a proxy entirely and has no genuine effect on the outcome Y itself? If there is a $S \to Y$, S itself might directly vary the treatment effect. If not, then all observed difference is due to $\mathbf{X}$.

For a simple numerical example of observing heterogeneity when there is truly no effect modification, i.e. S is merely a surrogate effect modifier, consider a binary treatment T, a binary effect modifier S, and a continuous outcome Y. Assume that the treatment effect depends strongly on a binary covariate X, so that $\mathbb{E}[Y(1) - Y(0) \mid X = 1] = 10$, but $\mathbb{E}[Y(1) - Y(0) \mid X = 0] = 0$. Also assume that the distribution of X across values of S is imbalanced: $\Pr(X = 1 \mid S = 1) = 0.8$, and $\Pr(X = 1 \mid S = 0) = 0.2$. Comparing the difference in CATEs, we end up with what looks like strong effect modification by S:

$$\mathbb{E}[Y(1) - Y(0) \mid S = 1] - \mathbb{E}[Y(1) - Y(0) \mid S = 0] = (0.8 \times 10) - (0.2 \times 10) = 8 - 2 = 6.$$

Conditional on X, however, $\mathbb{E}[Y(1) - Y(0) \mid S = 1, X] = \mathbb{E}[Y(1) - Y(0) \mid S = 0, X] = 10X$. Marginally, it looks like there is strong effect modification by S, but truly the observed difference is entirely due to X: effect heterogeneity can exist even when true modification is 0, meaning S is merely a surrogate effect modifier.

[7]There already exists literature on what is essentially the same estimand which calls the phenomenon "causal treatment effect moderation" (Bansak, 2021), but I decided to stick with well-established terminology on effect dependence instead. It is worth noting at this point though that Bansak (2021)'s "effect moderation" is effectively the same as causal effect modification, as I will show later.

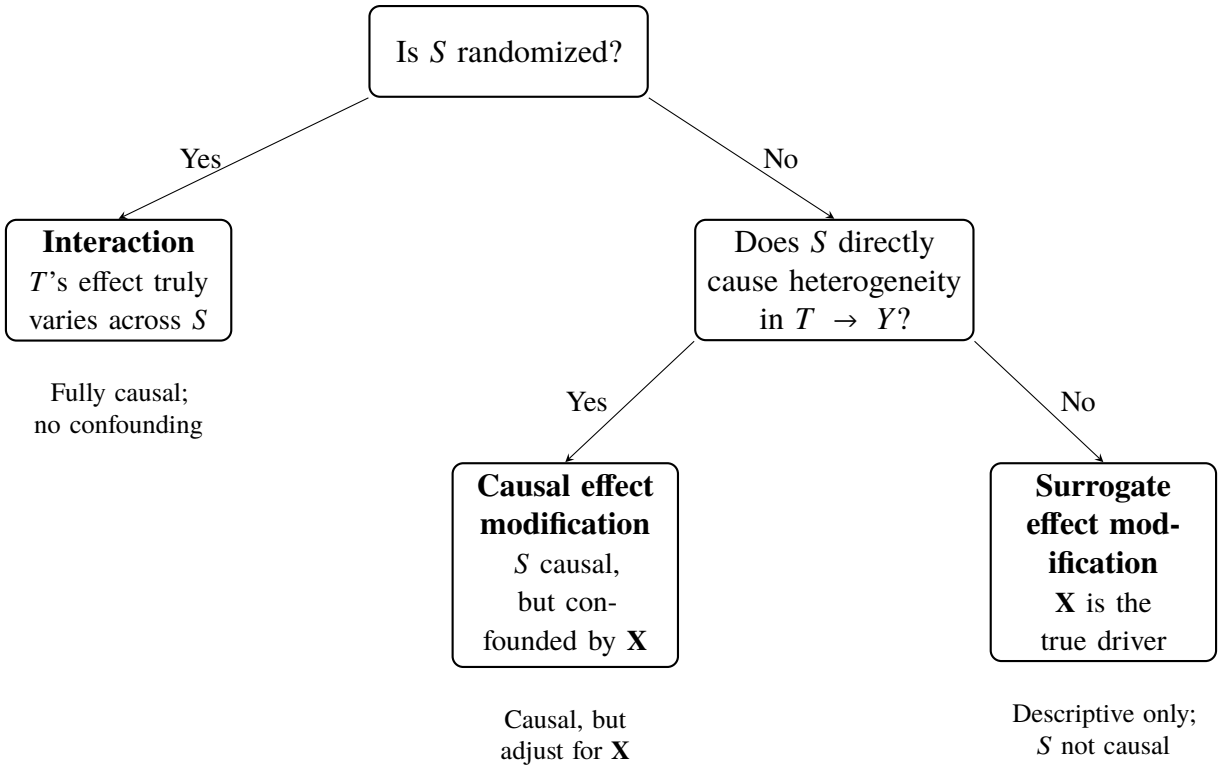


Figure 8: Types of effect dependency, when the treatment T is randomized by design.

Now that causal effect modification has been clarified conceptually, I will turn to the special case of the conjoint experiment, a factorial design commonly used in recent experimental political science research. I will show how the conjoint is a special case of the randomized controlled trial, and its quantities of interest are in principle equivalent to the ones discussed earlier. As such, effect dependency or heterogeneity faces the same identification challenges as any other experimental design.

2.4 Conjoint Experiments

Conjoint experiments present respondents with hypothetical profiles that vary simultaneously across multiple attributes, asking them to choose between or rate these profiles. By randomly assigning attribute values across profiles, conjoint designs exploit the logic of randomization to identify the causal effect of each attribute on choices or ratings. This makes them particularly well-suited to studying social judgments and decision-making, where multiple characteristics of a target person or object are likely to jointly influence behavior. A further advantage is that preferences are revealed implicitly through choices rather than stated directly, which reduces social desirability bias in the measurement of sensitive attributes such as race or gender (Hainmueller et al., 2014). The effects of individual attributes are then recovered by averaging over the random variation in all other attributes; a logic I formalize below.

Please read the descriptions of the potential immigrants carefully. Then, please indicate which of the two immigrants you would personally prefer to see admitted to the United States.

	Immigrant 1	Immigrant 2
Prior Trips to the U.S.	Entered the U.S. once before on a tourist visa	Entered the U.S. once before on a tourist visa
Reason for Application	Reunite with family members already in U.S.	Reunite with family members already in U.S.
Country of Origin	Mexico	Iraq
Language Skills	During admission interview, this applicant spoke fluent English	During admission interview, this applicant spoke fluent English
Profession	Child care provider	Teacher
Job Experience	One to two years of job training and experience	Three to five years of job training and experience
Employment Plans	Does not have a contract with a U.S. employer but has done job interviews	Will look for work after arriving in the U.S.
Education Level	Equivalent to completing two years of college in the U.S.	Equivalent to completing a college degree in the U.S.
Gender	Female	Male

	Immigrant 1	Immigrant 2
If you had to choose between them, which of these two immigrants should be given priority to come to the United States to live?	○	○

Figure 9: The Immigrant experiment: respondents have to choose which immigrant they would rather see admitted, depending on a number of randomized attributes.

Figure 9 shows an example choice task from Hainmueller et al. (2014), in which respondents were asked which of two immigrant profiles they would prefer to see admitted to the United States. Each profile is described by a number of different attributes, including country of origin, education, and gender, whose values are randomly assigned independently across profiles. A respondent who prefers Immigrant 2 in this task may be responding to the difference in job experience, education, country of origin, gender, or some combination of these; conjoints allow researchers to recover the average contribution of each attribute separately by aggregating over many such tasks.

Formally, let some attribute A_k take levels $\mathcal{A}_k = \{a_1, \dots, a_J\}$. In a conjoint experiment, we simultaneously randomize multiple attributes $\mathbf{A}$, and for any value $a \in \mathcal{A}_k$ we can define potential outcomes $Y(a, \mathbf{A}_{-k})$, where $\mathbf{A}_{-k}$ denotes all attributes except A_k. The fundamental quantity of interest in much of conjoint analysis is the Average Marginal Component Effect (AMCE, Hainmueller et al. (2014)), defined as the expected change in outcome when switching attribute A_k from a baseline value a_0 to some other value a', averaging over the experimental distribution of all remaining attributes:

$$\text{AMCE}(a') = \mathbb{E}[Y(a', \mathbf{A}_{-k}) - Y(a_0, \mathbf{A}_{-k})]$$

A closely related quantity is the marginal mean (Leeper et al., 2020), which is simply the expected

outcome when $A_k = a$, marginalizing over all other attributes:

$$\mathscr{M}(a) = \mathbb{E}[Y(a, \mathbf{A}_{-k})]$$

The AMCE is then a difference of marginal means: $\text{AMCE}(a') = \mathscr{M}(a') - \mathscr{M}(a_0)$. Because attribute values are randomized independently, each attribute is orthogonal to all others by design, which implies $A_k \perp\!\!\!\perp \mathbf{A}_{-k}$ and therefore $\mathbb{E}[Y(a, \mathbf{A}_{-k})] = \mathbb{E}[Y(a)]$. This means that for any single attribute, the conjoint is equivalent to a standard randomized experiment in which that attribute alone is the treatment and all other attributes average out as background noise. In the binary case, where $A_k = 1$ if the profile has level a' and $A_k = 0$ otherwise, this lets us restate the AMCE directly as an ATE:

$$\mathbb{E}[Y(a', \mathbf{A}_{-k}) - Y(a_0, \mathbf{A}_{-k})] = \mathbb{E}[Y(1) - Y(0)]$$

The AMCE for any attribute level is therefore simply the ATE of that level as a treatment, identified by the same randomization logic as a standard randomized controlled trial. On a sidenote, for the design to work, usually multiple subjects $i = \{1, \dots, N\}$ have to sit through multiple rounds $j = \{1, \dots, M\}$ of the experiment, so any single outcome is best thought of as $Y_{ij}(a'_k, \mathbf{A}_{-k})$, so our expectation of the outcome for a certain individual in a certain round. Hainmueller et al. (2014) show how estimates of the AMCE can easily be retrieved from a regression model, where we regress the outcome variable (e.g. whether a profile was selected or not) on attribute value dummies. The coefficients on the dummies are then the AMCE of the respective attribute value they represent.

In conjoint experiments, researchers might want to assess two types of effect dependency. First, they might be interested in whether the effect of one attribute A differs according to another attribute B. Omitting the other attributes $\mathbf{A} \setminus \{A, B\}$ for notational clarity, we can define the AIE for some treatment combination (a', b') relative to some baseline (a_0, b_0) following Egami and Imai (2018) as

$$\mathbb{E}[Y(a', b')] - \mathbb{E}[Y(a_0, b')] - \mathbb{E}[Y(a', b_0)] + \mathbb{E}[Y(a_0, b_0)]$$

which is an equivalent of the AIE as shown in eq. 2. Identification of this quantity is warranted by the design, as all attribute values are fully randomized: since the separate effects are identified, so are the joint effects, as layed out further above. The second kind of effect dependency of interest is what Leeper et al. (2020) call "subgroup analysis": whether the effect of an attribute differs according to some respondent attribute S:

$$\mathbb{E}[Y(a') - Y(a_0) \mid S = s]$$

The equivalence to CATEs (cf. eq. 1) in the case where A and S are binary (or at least dichotomizable) is immediately obvious. As such, if researchers would like to claim that S is actually the cause of any observed heterogeneity, they are plagued by the exact same identification problem of causal effect modification layed out above.

After reviewing the logic of the conjoint experiment and establishing that it faces the very same general identification challenges identified previously, I will now turn to outlining the proposed method for retrieving estimates of causal effect modification, even when the effect modifier of interest was not intervened on & is thus observational. I will start by defining the estimand, before describing the estimation procedure in more detail.

3 Method

Having clarified the conceptual distinction between effect modification and causal interaction, and how they also apply to conjoint experiments, I now turn to the estimation problem. While the approach proposed here applies to experimental designs generally, conjoint experiments are a particularly natural application context: the treatment of interest is randomized by design, while the respondent characteristics that researchers use to define subgroups of interest typically are not. This section first defines the estimand of interest and then introduces a weighting-based approach that allows causal effect modification to be estimated even when the effect modifier is not randomized.

3.1 Estimand

Recall that treatment effect heterogeneity by S refers to differences in the average treatment effect across levels of S. When S is not randomized, such differences cannot automatically be interpreted as causal effect modification because factors $\mathbf{X}$ may influence both selection into S and the response to treatment. In that case, differences in conditional treatment effects across levels of S may reflect covariate imbalance rather than genuine causal interaction. In an ideal setting, where both T and S could be experimentally manipulated, causal effect modification could be identified directly as causal interaction, by comparing the effect $T \to Y$ across values of S. Let

$$\delta_{\text{causal}}(s) = \mathbb{E}[Y(1, s) - Y(0, s)]$$

denote the causal effect $T \to Y$ when S is set to s. Causal effect modification $\Delta(S)$ is then analogous to interaction, as the difference in $T \to Y$ across values of S

$$\Delta(S)_{\text{causal}} = \delta_{\text{causal}}(1) - \delta_{\text{causal}}(0)$$

which can be estimated in terms of observed data as eq. 3 or 4. In the setting considered here, however, only the treatment T is randomized, while the effect modifier S is observational. We can still estimate CATEs

$$\mathbb{E}[Y(1) - Y(0) \mid S = s]$$

which describe how the treatment effect varies across observed levels of S. However, these quan-

tities do not have a causal interpretation with respect to S, since the distribution of pre-treatment covariates $\mathbf{X}$ may differ across levels of S. To recover causal effect modification, we must therefore adjust for covariates $\mathbf{X}$ that jointly affect selection into S and potential outcomes. Using the law of total expectation $\mathbb{E}[A] = \mathbb{E}_B[\mathbb{E}[A \mid B]]$, we can insert $\mathbf{X}$ as

$$\mathbb{E}[Y(t) \mid S = s] = \mathbb{E}_{\mathbf{X}}\Big[\mathbb{E}[Y(t) \mid S = s, \mathbf{X}]\Big].$$

so that

$$\delta(s) = \mathbb{E}_{\mathbf{X}}\Big[\mathbb{E}[Y(1) - Y(0) \mid S = s, \mathbf{X}]\Big].$$

This expression averages treatment effects within strata defined by both S and $\mathbf{X}$, and then integrates over the distribution of covariates. The estimand of interest is the difference in these adjusted treatment effects across levels of S:

$$\begin{aligned}\Delta(S) &= \delta(1) - \delta(0) \\ &= \mathbb{E}_{\mathbf{X}}\Big[\mathbb{E}[Y(1) - Y(0) \mid S = 1, \mathbf{X}]\Big] - \mathbb{E}_{\mathbf{X}}\Big[\mathbb{E}[Y(1) - Y(0) \mid S = 0, \mathbf{X}]\Big]\end{aligned} \tag{6}$$

Under appropriate identification assumptions, this quantity can be written in terms of observed data as

$$\begin{aligned}&\mathbb{E}_{\mathbf{X}}\Big[\mathbb{E}[Y \mid T = 1, S = 1, \mathbf{X}] - \mathbb{E}[Y \mid T = 0, S = 1, \mathbf{X}]\Big] \\ &-\mathbb{E}_{\mathbf{X}}\Big[\mathbb{E}[Y \mid T = 1, S = 0, \mathbf{X}] - \mathbb{E}[Y \mid T = 0, S = 0, \mathbf{X}]\Big]\end{aligned} \tag{7}$$

This estimand captures how the causal effect of the treatment changes across values of the effect modifier after adjusting for differences in covariate distributions. To interpret this quantity causally, several assumptions are required. Since S is non-random, and we know that factors $\mathbf{X}$ exist, we know the independence assumption neccessary for recovering causal interaction effects to be violated. To make the difference in ATEs across S causal, we thus have to assume first conditional ignorability of the effect modifier given pre-treatment covariates $\mathbf{X}$.

Assumption 1. *Conditional Independence for the effect modifier.*

$$(Y(1), Y(0)) \perp\!\!\!\perp S \mid \mathbf{X}$$

Conditional on $\mathbf{X}$, the potential outcomes are independent of S. This implies that there is no unobserved or unaccounted for variable next to $\mathbf{X}$ that affects selection into groups of S and varies the treatment effect, meaning no hidden confounding after conditioning on $\mathbf{X}$.

We also have to make some additional assumptions, which are mostly the usual assumptions we have

to make to estimate ATEs, with some modifications given the conditional nature, and the nature of S.

Assumption 2. *Independence (complete randomization) of the treatment.*

$$(Y(1), Y(0)) \perp\!\!\!\perp T_i$$

The potential outcomes are independent of the treatment, meaning there are no factors affecting the potential outcomes and selection into treatment or control groups.

Next, for each value of the covariates, there has to be a nonzero probability of being either group (or have either value) of S, and that there is no covariate value or combination of values that make it impossible to be in either group or have either value, e.g. due to logical constraints.

Assumption 3. *Positivity for the modifier.*

$$0 < \Pr(S = 1 \mid \mathbf{x}) < 1 \ \forall \mathbf{x} \in \mathscr{X}$$

For the treatment variable, this is warranted by complete randomization. Note that the complete randomization of the treatment also ensures $0 < \Pr(T_i = t, S_i = s) < 1 \ \forall i, t \in \mathscr{T}, s \in \mathscr{S}$ i.e. all combinations of treatment and effect modifier values are possible for all units.

Assumption 4. *Stable Unit Treatment Values (SUTVA) for treatment and effect modifer.* The potential outcomes for each subject do not depend on the effect modifier value or treatment status of other units. Concretely, this means no carryover or spillover effects, and no interference across subjects: whether subject i was treated or not, or what value of the effect modifier it has, should not affect the outcomes of any other subject.

Assumption 5. *Pretreatment effect modifier and confounders.* The effect modifier S and the set of confounding variables $\mathbf{X}$ are assumed to be pretreatment, such that the treatment T can have no individual-level effect on S or $\mathbf{X}$, meaning S, $\mathbf{X}$, and the potential outcomes are jointly independent of the treatment (Bansak, 2021). This also makes causal effect modification conceptually distinct from causal interaction: non-randomization of S forces us to assume a fixed temporal ordering of variables or measurements.

Under assumptions 1-5, the estimand from eq. 6 becomes estimable in terms of observed data as eq. 7. This estimand derived from CATEs, starting from non-causal effect modification or treatment effect heterogeneity, can easily be restated as the average treatment moderation effect (ATME, Bansak, 2021):

$$\begin{aligned} \mathbb{E}_{\mathbf{X}}\Big[& \mathbb{E}[Y \mid T = 1, S = 1, \mathbf{X}] - \mathbb{E}[Y \mid T = 0, S = 1, \mathbf{X}] \\ & - \mathbb{E}[Y \mid T = 1, S = 0, \mathbf{X}] + \mathbb{E}[Y \mid T = 0, S = 0, \mathbf{X}]\Big] \end{aligned} \tag{8}$$

This quantity is a conditional version of causal interaction, reducing to the additive interaction effect in equations 3 and 4 when both T and S are randomized, and conditioning on $\mathbf{X}$ is therefore

not needed. Thus, effect modification, causal and non-causal, and additive causal interaction are fundamentally the same estimand expressed under different identifying conditions; they differ primarily in whether adjustment is required for the distribution of the effect modifier. The substantive contribution of this thesis is therefore not the introduction of a new causal estimand, but a method for recovering a familiar interaction quantity under weaker design conditions. The estimand, and other restatements of the same quantity such as the ATME, are simply a conditional version of the AIE.

Now that I have defined causal effect modification, and clarified the estimand of interest, I will go into how to retrieve estimates of causal effect modification. I will start from linear models with interaction terms, which are commonly used in the social sciences, before suggesting a better estimand based on inverse probability weighting and flexible propensity score estimation that is also consistent under nonlinearity of selection into S across $\mathbf{X}$, or nonlinear treatment effect heterogeneity across $\mathbf{X}$.

3.2 Estimation With Linear Models & Estimation Challenges

Starting from assessing effect heterogeneity, researchers commonly fit a simple linear model with an interaction term for $T \times S$:

$$Y = \beta_0 + \beta_1 T + \beta_2 S + \beta_3 (T \times S) + \varepsilon$$

To see how the coefficient on the interaction term β_3 corresponds to effect modification (or heterogeneity) across S (cf. eq. 5), we can decompose the model (for a binary T and S):

$$\begin{aligned}
\tau(1) &= \mathbb{E}[Y \mid T = 1, S = 1] - \mathbb{E}[Y \mid T = 0, S = 1] \\
&= (\beta_0 + \beta_1 + \beta_2 + \beta_3) - (\beta_0 + \beta_2) \\
&= \beta_1 + \beta_3, \\
\tau(0) &= \mathbb{E}[Y \mid T = 1, S = 0] - \mathbb{E}[Y \mid T = 0, S = 0] \\
&= (\beta_0 + \beta_1) - \beta_0 \\
&= \beta_1
\end{aligned}$$

so that

$$\Delta_{\text{CATE}}(S) = \tau(1) - \tau(0) = (\beta_1 + \beta_3) - \beta_1 = \beta_3.$$

This also makes it obvious that β_3 is not a causal estimate for effect modification by S, since we never account for $\mathbf{X}$. While papers advising this never claim that β_3 is causal effect modification, they also frequently make little effort to disentangle causal from non-causal effect modification (cf. e.g. Christensen et al., 2021; Leeper et al., 2020). The simplest way of accounting for $\mathbf{X}$, as used or advised by some researchers (Dong, 2015; Lee & Lee, 2025), is

$$Y = \beta_0 + \beta_1 T + \beta_2 S + \beta_3 (T \times S) + \mathbf{X}^\top \boldsymbol{\gamma} + \varepsilon$$

However, by not modelling heterogeneity in effects attributable to $\mathbf{X}$ via an interaction term $T\mathbf{X}^\top \boldsymbol{\gamma}_2$, this model incorrectly forces the treatment effect to be constant across $\mathbf{X}$. Including $\mathbf{X}^\top \boldsymbol{\gamma}_1$ accounts for differences in outcomes across $\mathbf{X}$, while $T\mathbf{X}^\top \boldsymbol{\gamma}_2$ accounts for differences in treatment effects across $\mathbf{X}$. Since omitting the latter term incorrectly forces the treatment to be constant across $\mathbf{X}$, heterogeneity in treatment effects due to $\mathbf{X}$ might mistakenly be attributed to S, if $\mathbf{X}$ and S are correlated, biasing β_3. So heterogeneity caused by $\mathbf{X}$ should explicitly be modelled as[8]

$$Y = \beta_0 + \beta_1 T + \beta_2 S + \beta_3 (T \times S) + \mathbf{X}^\top \boldsymbol{\gamma}_1 + T\mathbf{X}^\top \boldsymbol{\gamma}_2 + \varepsilon.$$

This linear model delivers a consistent causal estimate of β_3, if modeling assumptions hold, as it also explicitly models $\mathbf{X}$'s interaction with the treatment. To see that β_3 in this model is an estimate of $\Delta(S)$ (cf. eq. 6), we can decompose it the same way as before:

$$\begin{aligned}
\delta(1) &= \mathbb{E}[Y \mid T = 1, S = 1, \mathbf{X}] - \mathbb{E}[Y \mid T = 0, S = 1, \mathbf{X}] \\
&= (\beta_0 + \beta_1 + \beta_2 + \beta_3 + \mathbf{X}^\top \boldsymbol{\gamma}_1 + \mathbf{X}^\top \boldsymbol{\gamma}_2) - (\beta_0 + \beta_2 + \mathbf{X}^\top \boldsymbol{\gamma}_1) \\
&= \beta_1 + \beta_3 + \mathbf{X}^\top \boldsymbol{\gamma}_2, \\
\delta(0) &= \mathbb{E}[Y \mid T = 1, S = 0, \mathbf{X}] - \mathbb{E}[Y \mid T = 0, S = 0, \mathbf{X}] \\
&= (\beta_0 + \beta_1 + \mathbf{X}^\top \boldsymbol{\gamma}_1 + \mathbf{X}^\top \boldsymbol{\gamma}_2) - (\beta_0 + \mathbf{X}^\top \boldsymbol{\gamma}_1) \\
&= \beta_1 + \mathbf{X}^\top \boldsymbol{\gamma}_2;
\end{aligned}$$

so that

$$\begin{aligned}
\Delta(S) &= \delta(1) - \delta(0) \\
&= (\beta_1 + \beta_3 + \mathbf{X}^\top \boldsymbol{\gamma}_2) - (\beta_1 + \mathbf{X}^\top \boldsymbol{\gamma}_2) \\
&= \beta_3.
\end{aligned}$$

This also corresponds to fitting separate regressions for treated and control units: as Bansak (2021) shows, this is equivalent to splitting the data by treatment assignment, regressing Y on S separately, and then taking the diferrence in effects as an estimate of causal moderation:

$$\begin{aligned}
Y_i &= \alpha_0 + \alpha_1 S_i + \mathbf{X}_i^\top \boldsymbol{\gamma} + \varepsilon_i, \ \forall i : T_i = 1 \\
Y_i &= \beta_0 + \beta_1 S_i + \mathbf{X}_i^\top \boldsymbol{\zeta} + \varepsilon_i, \ \forall i : T_i = 0 \\
\Delta &= \alpha_1 - \beta_1
\end{aligned} \tag{9}$$

Intuitively, this owes to the fact that, once separate effects are identified, interaction is symmetric, and the difference in $S \to Y$ across T is the same as the difference in $T \to Y$ across S. To see why

[8] I illustrate this point in the simulation study in the appendix.

that is, refer to section 2.2.2, specifically eqs. 3 & 4

The main limitation of the linear model is that it imposes restrictive functional form assumptions on all relationships in the model. However, such assumptions are unlikely to hold in many social science settings. In the specific setting I consider, a randomized treatment and a nonrandomized effect modifier, violations of the linearity assumption might arise in two different places: first, selection into levels of S may occur as a nonlinear function of $\mathbf{X}$, for example via thresholds, or interactions between covariates. Second, the treatment effect may vary nonlinearly across $\mathbf{X}$.[9]

Under these conditions, estimation strategies that rely on rigid assumptions, and on the researcher to correctly specify selection and heterogeneity dynamics, may be sensitive to model misspecification and functional form dependence. To address this problem, flexible modelling strategies can be used to approximate complex assignment mechanisms and heterogeneous response structures without imposing strong assumptions. The approach proposed in this thesis follows this logic by using inverse probability weighting as a preprocessing step, combined with a simple outcome model to estimate causal effect modification while reducing model dependence. While fully nonparametric approaches, for example using targeted maximum likelihood estimation (Bodnar et al., 2022), could further relax modelling assumptions, the strategy adopted here occupies a middle ground: weighting is used to adjust to balance the distribution of $\mathbf{X}$ across S, while the quantity of interest is still estimated using a simple and interpretable parametric model. It is best to think of weighting as "nonparametric preprocessing" (Ho et al., 2007): we first create a pseudopopulation in which $S \perp\!\!\!\perp \mathbf{X}$, and then estimate a parametric model on *those* data.

3.3 Estimation Strategy

If $\mathbf{X}$ affects both selection into S, and varies the treatment effect, different groups of S will differ in their distribution of $\mathbf{X}$, and differences in outcomes across S may simply be due to difference in $\mathbf{X}$ across S. Heterogeneity in treatment effects truly due to $\mathbf{X}$ will mistakenly be attributed to S. The idea is to preprocess the data such that, in the preprocessed data, $S \perp\!\!\!\perp \mathbf{X}$. In other words, we want to balance the data so that the distribution of $\mathbf{X}$ becomes the same across groups of S. Since the distribution of $\mathbf{X}$ is equal across S, so is the distribution of effect heterogeneity caused by $\mathbf{X}$.

I propose to use inverse probability weighting in a three-step approach. First, we estimate the probability of being in the $S = 1$ group given $\mathbf{X}$, the so-called propensity score, using flexible, non-parametric models.[10] Second, units are weighted by the inverse of this conditional probability, giving more weight to observations that are underrepresented given their covariate values; this is known as "inverse probability weighting" (Robins et al., 1994, 2000). Third, a simple outcome model, such as linear regression, is fit to the weighted data. Crucially, this approach retains the main advantage of linear regression: the outcome model remains easily interpretable, while weighting

[9] Hainmueller et al. (2019) show that the linearity assumption is especially unlikely to hold for multiplicative interaction terms.

[10] As I will elaborate on further below, it should really say *potentially* using flexible models, since researchers should optimally choose the propensity score model that achieves the best covariate balance. If it turns out a parametric model, such as logistic regression, delivers better covariate balance, there is no reason to use a more flexible model that performs worse.

accounts for potentially complex, nonlinear selection mechanisms.[11] Nonparametric machine learning (ML) models, such as for example bagged or boosted tree-based learners, are well-suited for estimating propensity scores because they can automatically capture complex relationships such as nonlinearities, threshold effects, or interactions among covariates, relieving researchers of the burden to explicitly model these relationships in parametric regression models.

Below, I begin by outlining inverse probability weighting. I then turn to using ML models for the estimation of propensity scores, where I mostly deal with finding the model configuration that produces the best resulting covariate balance.

3.3.1 Inverse Probability Weighting

Inverse probability weighting (IPW) (Robins et al., 1994, 2000) is used to adjust for confounding in observational studies, where the treatment is not randomly distributed. IPW uses the propensity score to adjust for covariate imbalance between the treatment and control group, by weighting each subject by the inverse of its probability of receiving the treatment. Intuitively, we can adapt and use this to create a pseudopopulation in which the distribution of covariates is balanced across groups of S, allowing differences in treatment effects across these groups to be attributed to S itself rather than to confounding differences in $\mathbf{X}$. I will start by explaining the basic idea of IPW before elaborating on how to adapt it to the problem at hand.

The propensity score π is defined as the conditional probability of treatment given covariates

$$\pi_T(\mathbf{X}) = \Pr(T = 1 \mid \mathbf{X})$$

or, in other words, "the propensity toward exposure to treatment (…) given the observed covariates" $\mathbf{X}$ (Rosenbaum & Rubin, 1983, p. 42). Rosenbaum and Rubin (1983) show that adjusting solely for differences between treated and control units in the propensity score removes all biases associated with differences in covariates (cf. also Hirano et al. (2003)). Their key insight was that if treatment and potential outcomes are independent conditional on all covariates, they are also independent conditional on the conditional probability of receiving treatment given covariates. Formally, as shown by Rosenbaum and Rubin (1983), unconfoundedness implies $T \perp\!\!\!\perp (Y(1), Y(0)) \mid \pi_T(\mathbf{X})$. Hirano et al. (2003) show that weighting by the inverse of a nonparametric estimate of the propensity score yields an efficient estimate of the average treatment effect (ATE).

Conceptually for a treatment T, in a randomized trial, $T \perp\!\!\!\perp \mathbf{X}$ by design, which means $\Pr(T = 1 \mid \mathbf{X}) = \Pr(T = 1)$. In observational data, treatment assignment is not independent of $\mathbf{X}$, s.t. $\Pr(T = 1 \mid \mathbf{X}) \neq \Pr(T = 1)$. As such, any effect of a treatment T on an outcome Y is potentially subject to confounding from $\mathbf{X}$, and observed changes in Y with respect to T might really be attributable to imbalances in $\mathbf{X}$ across values of T. IPW addresses this by upweighting underrepresented treatment-covariate combinations, and downweighting overrepresented ones, balancing the distribution of $\mathbf{X}$ across treatment and control group, s.t. $\Pr_w(T = 1 \mid \mathbf{X}) = \Pr(T = 1)$.

In the setting at hand, T is fully randomized, and we seek to balance the distribution of $\mathbf{X}$ across values of S, so we estimate propensity for S instead:

[11] I briefly address the point why weighting is preferrable to matching in the next section.

$$\pi_S(\mathbf{X}) = \Pr(S = 1 \mid \mathbf{X})$$

Weights are then calculated as

$$w_i = \begin{cases} \frac{1}{\pi_S(X_i)} & \text{if } S_i = 1 \\ \frac{1}{1-\pi_S(X_i)} & \text{if } S_i = 0 \end{cases}$$

One important edge case to consider are extreme weights. In cases of strong imbalance, for example when some covariate combinations are very predictive of S-values, subjects might end up with very high or very low probability of $S = 1$, which might cause extreme weight values, inflating variance and confidence intervals of effect estimates (Chesnaye et al., 2022; Cole & Hernán, 2008). We can deal with this by truncating extreme weights, and/or stabilizing (Cole & Hernán, 2008):

$$w_i = \begin{cases} \frac{\Pr(S=1)}{\pi_S(X_i)} & \text{if } S_i = 1 \\ \frac{\Pr(S=0)}{1-\pi_S(X_i)} & \text{if } S_i = 0 \end{cases}$$

Here, we simply replace the numerator with the unconditional probabilities of $S = 1$ or $S = 0$. As weighting introduces within-subject correlation in outcomes in the weighted sample, which needs to be accounted for by using robust variance estimators or bootstrap-based methods (Austin, 2016; Chesnaye et al., 2022). Figure 10 shows intuitively how weighting creates a pseudopopulation where the distribution of covariates is balanced across groups of S.

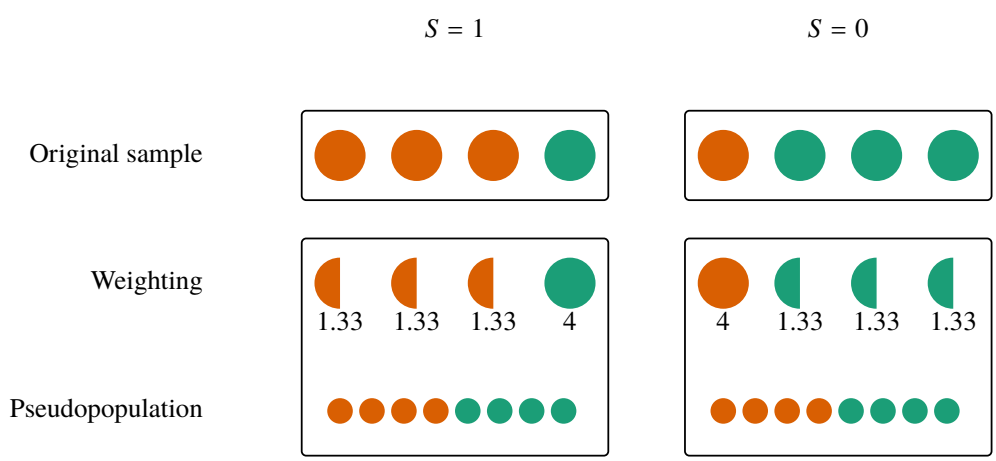


Figure 10: Covariate imbalance across groups of S. A red circle means the subject has value $X = 1$, and a green circle means the subject has $X = 0$. $\Pr(S = 1 \mid X = 1) = 0.75$, resulting in a weight of $\frac{1}{0.75} = 1.33$, while $\Pr(S = 1 \mid X = 0) = 0.25$, with a weight of $\frac{1}{0.25} = 4$. Conversely, $\Pr(S = 0 \mid X = 1) = 0.25$ yielding a weight of $\frac{1}{0.25} = 4$ again, and $\Pr(S = 0 \mid X = 0) = 0.75$ with a weight of again $\frac{1}{0.75} = 1.33$. By applying the weights we create a pseudopopulation where $X = 1$ is distributed evenly across groups with a prevalence of 50%.

Adjustment on π_S removes bias associated with imbalance in $\mathbf{X}$ across values of S (Rosenbaum

& Rubin, 1983). To use the weights for adjustment, we can use them in a weighted least squares regression

$$\arg\min_{\boldsymbol{\beta}} \sum_i w_i (Y_i - f(T_i, S_i, \mathbf{X}_i, \boldsymbol{\beta}))^2$$

where β_3 would be the estimate of causal effect modification by S, and $f(T_i, S_i, \mathbf{X}_i, \boldsymbol{\beta})$ is the specification to be fit. In theory, one could also match $S = 1$ units to $S = 0$ units on the estimated propensity score (Abadie & Imbens, 2016; Rosenbaum & Rubin, 1983), but without further adjustment, propensity score matching (PSM) yields not an adjusted CATE estimate, but rather estimates an adjusted conditional version of the average treatment effect among treated units (ATT, or perhaps rather CATT in this setting). If we fail to find proper matches for all $S = 1$ units, the interpretation of the final matching estimator changes even further.[12] The core issue in this procedure is the estimation of π_S, which might become tricky especially when there are many covariates in $\mathbf{X}$, and they affect selection into groups of S in non-linear, or otherwise complex ways. One potential remedy is to use flexible models to estimate π_S.

3.3.2 Estimating Propensity Scores Using ML

The key step in inverse probability weighting is the estimation of the propensity scores, the conditional probability $\Pr(S = 1 \mid \mathbf{X})$. Estimating this quantity is key to IPW because the resulting weights are only valid if they achieve balance in the covariates $\mathbf{X}$ across S.

In many social science settings, selection into S may depend on complex relationships among covariates, such as nonlinearities, interactions, or threshold effects. Despite this, researchers still mostly resort to logistic regression to estimate propensity scores (Leite et al., 2025). Standard logistic regression assumes linearity on the logit scale and requires explicit modeling of interactions or nonlinearities. When using parametric models to estimate propensity scores, misspecifications of the models or violations of functional form assumptions means covariate balance might not be achieved when conditioning on the estimated propensity scores (Drake, 1993). This means that if the researchers fails to correctly specify nonlinearity or otherwise complex relationships, such as interactions between covariates, the final estimate will again be biased. A natural solution is to use more flexible models that can automatically capture complex patterns in the data. Tree-based methods, such as gradient boosting machines (GBM), are particularly well-suited: they can dynamically learn relationships like nonlinearities, threshold effects, and interactions without explicit specification, are relatively insensitive to scale, and can accommodate categorical features (cf. Hastie et al., 2009; McCaffrey et al., 2004).

When actually fitting models for propensity score estimation, researchers need to be aware that most models will optimize for predictive performance. However, somewhat unintuitively, conditioning on the true propensity score does not neccessarily yield the best balance (Hirano et al., 2003; Imai & Ratkovic, 2014), and we thus want to find the estimate of the propensity score that optimizes covariate balance. Approaches explicitly targeting balance, such as the covariate balancing propen-

[12] PSM has also been drawn into doubt more generally (cf. King & Nielsen, 2019), but that is far beyond the scope of this thesis.

sity score (CBPS) (Imai & Ratkovic, 2014) are currently limited to parametric models. In practice, a pragmatic alternative is to perform hyperparameter[13] tuning for covariate balance when using ML models. While I do not claim any formal guarantees, and there are no formal evaluations of this procedure that I am aware of, it neccessarily produces better balancing estimates than optimizing purely for predictive performance.

Generally, there is no one-size-fits-all solution to estimating propensity scores, as the best model is dependent on the data. If ML models are used, the hyperparamters should be tuned, as the optimal choice of hyperparameters is highly data specific (Bergstra & Bengio, 2012; Boehmke & Greenwell, 2020): choosing hyperparameters based on what other researchers use, or what literature suggests, should be viewed with the greatest suspicion.

First, we need a consistent definition of covariate imbalance. A common approach is to consider (absolute) standardized mean differences (SMDs) between covariates, where a difference in SMDs $< |0.1|$ is considered negligable imbalance (Austin, 2009, 2019; Cohen, 1988; Mamdani et al., 2005). Using, for example, Cohen's d, the standardized difference between two groups $S = 1$ and $S = 0$ for a covariate X_j is

$$d(X_j) = \frac{\bar{X}_j^{S=1} - \bar{X}_j^{S=0}}{\sigma_j^{\text{Pooled}}}$$

where σ_j^{Pooled} is the pooled standard deviation of X_j across the two groups. Suppose that we have a propensity score model m_ϕ with hyperparameters $\phi \in \mathbf{\Phi}$, producing estimated propensity scores:

$$\pi_i^\phi = m_\phi(\mathbf{X}_i), \quad i = 1, \dots, N.$$

Using these, we then calculate stabilized inverse probability weights for each unit:

$$w_i^\phi = \begin{cases} \frac{\Pr(S=1)}{\pi_i^\phi} & \text{if } S = 1 \\ \frac{\Pr(S=0)}{1-\pi_i^\phi} & \text{if } S = 0. \end{cases}$$

Weighted group means for X_j are computed as

$$\bar{X}_j^{S=s} = \frac{\sum_{i=1}^N w_i^\phi \mathbb{1}\{S_i = s\} X_{ij}}{\sum_{i=1}^N w_i^\phi \mathbb{1}\{S_i = s\}}, \; s \in \{0, 1\}.$$

We can then calculate Cohen's d as the standardized difference between weighted group means:

$$d_j^\phi = \frac{\bar{X}_j^{S=1,\phi} - \bar{X}_j^{S=0,\phi}}{\sigma_j^{\text{Pooled},\phi}}$$

[13] A hyperparameter is a parameter of the learning process, not of the model itself. As such, it is not affected by the learning process itself. It is set prior to, and remains constant during, training (cf. e.g. Géron, 2019)

Finally, to select the hyperparameters ϕ that produce the best covariate balance, we minimize the mean absolute SMD across all J covariates:

$$\mathscr{L}_{\text{Imbalance}}(\phi) = \frac{1}{J}\sum_{i=1}^{J} |d_j^{\phi}(X_j)|.$$

Hyperparameter tuning can be performed by searching over $\phi \in \mathbf{\Phi}$ to minimize $\mathscr{L}_{\text{Imbalance}}(\phi)$, producing a propensity score model that yields well-balanced covariates across S.

Usually, it is not feasible to evaluate the whole parameter space, and we can only evaluate some subset $\Phi \subset \mathbf{\Phi}$. A simple approach is regular grid search, where we evaluate all combinations of hyperparameter values in Φ. Alternatives include random grid search, which randomly samples configurations from $\mathbf{\Phi}$ and can be more efficient than regular grid search (Bergstra & Bengio, 2012), or iterative approaches like sequential model-based optimization/Bayesian optimization, which iteratively evaluate hyperparameter combinations based on past results (Hutter et al., 2011; Snoek et al., 2012). Algorithm 1 is a simple algorithm for tuning for covariate balance using grid search, for either regular or (pseudo-)random grids. We do not need to worry about cross-validating or overfitting in the traditional sense, as this is not a traditional prediction or inference problem, and we are not looking for a model that generalizes well: we are looking for the model that, given the data at hand, estimates $S = 1$ probabilities such that weighting observations on their inverse produces the best covariate balance between two groups.

Algorithm 1: Hyperparameter tuning for covariate balance using grid search.

Input: Hyperparameter candidates $\Phi \subset \mathbf{\Phi}$, propensity score model $m(\cdot)$, data $\{S_i, \mathbf{X}_i\}_{i=1}^{N}$
Output: Best model m_{ϕ}^{*} and hyperparameters ϕ^{*} minimizing covariate imbalance
foreach $\phi \in \Phi$ **do**
  Fit the propensity score model with hyperparameters ϕ: $\pi_i^{\phi} = m_{\phi}(\mathbf{X}_i)$ for $i = 1, \dots, N$;
  Compute stabilized weights w_i^{ϕ} based on π_i^{ϕ};
  Compute weighted covariate means and standardized mean differences d_j^{ϕ} for $j = 1, \dots, J$;
  Compute overall imbalance $\mathscr{L}_{\text{imbalance}}(\phi) = \frac{1}{J}\sum_{j=1}^{J} |d_j^{\phi}|$;
  Store $(\phi, \mathscr{L}_{\text{imbalance}}(\phi))$ in results;
end
Select the hyperparameters with minimal imbalance: $\phi^{*} = \arg\min_{\phi \in \Phi} \mathscr{L}_{\text{imbalance}}(\phi)$;
Set $m_{\phi}^{*} = m_{\phi^{*}}$;
return m_{ϕ}^{*}, ϕ^{*}

In summary, this approach to propensity score estimation prioritizes covariate balance over predictive accuracy, reflecting the specific demands of the IPW framework. Flexible ML models tuned against imbalance offer a principled way to recover balance even when selection into S follows complex patterns that simpler models may fail to capture, or that researchers may fail to specify properly.

One more thing to point out is that, as long as researchers are not interested in the structural form of the propensity score model, they should be agnostic with regard to the kind of modelling ap-

proach used for propensity score estimation. They should neither resort to methods using logistic regression simply because of familiarity, nor resort to ML models for their promise of capturing complexity: the best model is highly data specific and there might be situations where parametric models outperform nonparametric models in achieving the best balance. In practice, researchers should tune and fit multiple different models, and use the one that produces the best balance, as long as there are no special requirements or interests in the structural form of the propensity score model.

Once balance is achieved (e.g. operationalized as all SMDs falling below $|0.1|$), the resulting weights render S approximately independent of $\mathbf{X}$, so that remaining variation in the treatment effect $T \to Y$ can more plausibly be attributed to S itself rather than confounding by $\mathbf{X}$. In practice, however, *perfect* balance is unlikely to be achieved, meaning that there will almost always be some residual imbalance in $\mathbf{X}$ across S remaining. A simple difference-in-means estimator applied to the weighted data will therefore likely still be biased, and researchers should additionally fit a parametric outcome model that explicitly controls for $\mathbf{X}$, as shown further above. Weighting should therefore best be viewed as a preprocessing step meant to get rid of imbalance that the parametric model to be fit is unlikely to capture properly.

Bansak (2021) suggests *parallel estimation*: similar to eq. 9 for regression adjustment, he suggests splitting the data by treatment status T, then estimating the effect $S \to Y$ separately within treatment groups, adjusting for $\mathbf{X}$, and taking the difference in $S \to Y$ as estimate of causal effect modification, for other forms of adjustment as well. However, for the proposed IPW approach, that would mean tuning at least two separate models for the same task, which introduces overhead. I suggest pooling instead, i.e. not splitting the sample and targeting $S \to (T \to Y)$ directly. The intuition behind pooling is that randomization of T ensures that $T \perp\!\!\!\perp (S, \mathbf{X})$, meaning $\Pr(S = 1 \mid \mathbf{X}, T) = \Pr(S = 1 \mid \mathbf{X})$), so we can simply estimate one propensity score model $\pi_S(\mathbf{X})$ over the whole data and weight on the inverse of its estimates, and we do not need to split by treatment assignment. I validate this intuition in a simulation study in the appendix

Finally, it bears emphasizing that this procedure does not eliminate all threats to valid inference: it only addresses confounding through observed covariates, and any unobserved variable that both induces selection into S and varies the effect $T \to Y$ would remain a source of bias. Researchers should further keep in mind that this adjustment is specific to *one* effect modifier, and if different effect modifiers for different treatments are examined, weights should not be reused.

To show that this approach is consistent, I also include a simulation study in the appendix.

4 Case Study

To illustrate the proposed method and demonstrate its applicability in a real empirical setting, I present a case study examining whether exposure to conflict-related sexual violence (CRSV) causally modifies the effect of ethnic group membership on interpersonal trust. This setting is well-suited to the method for several reasons. The treatment of interest, ethnic group membership of a hypothetical target person, is randomized by design, as it is embedded in a conjoint experiment. The potential effect modifier, exposure to CRSV, is observational and cannot be randomized

for both practical and ethical reasons, making it a canonical instance of the problem this thesis addresses. Moreover, there are theoretically motivated reasons to expect both that CRSV exposure is confounded by background characteristics, and that those same characteristics plausibly shape intergroup trust, meaning that naive estimates of effect modification are unlikely to have a causal interpretation.

The case study proceeds in three steps. I first situate the analysis in the relevant substantive literature and derive theoretical expectations. I then describe the data, including the survey design, the measurement of CRSV exposure, and the conjoint experiment used to elicit trust behavior. Finally, I outline the estimation procedure and present results across a series of models of increasing credibility, comparing naive, controlled, weighted, and matched estimates of causal effect modification by CRSV.

4.1 Background & Theory

A substantial body of literature establishes that exposure to violence in conflict settings has profound effects on the attitudes of those affected. At the individual level, it fosters stress and heightened threat perception, encouraging exclusionary attitudes (Beber et al., 2014; Canetti-Nisim et al., 2009), and erodes both interpersonal and institutional trust (Conzo & Salustri, 2019; Hager et al., 2019; Rohner et al., 2013). These effects are not confined to direct victims: consistent with the broader expectation that trauma transmits across generations (Felsen, 1998), exposure to conflict-related violence (CRV) within the family has been shown to reduce prosocial behavior even among those not directly affected (Schaub, 2023).

However, findings differ on how the experience of CRV shapes in- and outgroup trust, and ingroup boundaries, with some arguing that conflict reinforces ethnic cleavages and ingroup identification (Rohner et al., 2013), and others arguing that the adverse effects apply to interpersonal trust more generally, and are similarly stark across groups (Hager et al., 2019). A meta-analysis of the effects of exposure to CRV has found it generally increases antagonism toward wartime enemies and threat perception, and lowers intergroup trust (Barceló, 2025).

CRSV is a distinct form of CRV (Nordås & Cohen, 2021), and its specific consequences for trust and prosocial behavior remain comparatively underexplored. To assess the effect of CRSV on ethnic inter- and intragroup trust, I draw on original data from Bosnia and Herzegovina, where the 1992–1995 war following the collapse of Yugoslavia exposed civilians to severe and widespread CRV at the hands of both combatants and other civilians (Kalyvas & Sambanis, 2005). CRSV in particular "was perpetrated across the conflict against both women and men, often in detention and rape camps, serving as an instrument of ethnic persecution and patriarchal domination" (Schaub et al., 2025). Given its explicit ethnic dimension, the Bosnian case offers a particularly informative setting in which to examine the downstream consequences of CRSV for post-conflict interethnic relationships and trust.[14]

[14]While violence along ethnic lines is sometimes cited as defining feature of the conflict, others point out that the ethnic divide was not as deep as the violence suggests, and that civilian-on-civilian violence often "hid personal grudges rather than impersonal ethnic animosity" (Kalyvas & Sambanis, 2005, p. 215).

4.2 Data

I use original survey data (N=2674) collected by a reputable survey company in face-to-face interviews (CAPI) in Bosnia and Herzegovina, from September to December 2024 (Schaub et al., 2025). The survey included questions on wartime victimization, trust, and an embedded conjoint experiment to measure identity-based trust and was based on a stratified multi-stage sampling procedure. In a first step, enumerator starting points were identified, followed by respondent selection using a random walk procedure. The primary sampling units were Bosnia and Herzegovina's 6,141 naselja (settlements), which represent the smallest administrative units in the country. From these, 162 PSUs were randomly selected with probabilities proportional to population size, based on population data from the Agency for Statistics of Bosnia and Herzegovina (ASBH). In the initial sampling stage, the population was stratified into areas with relatively higher and lower risks of sexual victimization, with the former deliberately oversampled. Within each selected settlement, enumerator starting points were determined by overlaying settlements with a 0.006-degree raster grid. Grid cells were then selected through an additional population-weighted random draw, with population estimates derived from the Global Human Settlement Layer (GHSL) developed by the European Commission's Joint Research Centre (European Commission, 2023). Starting points were roadside locations within the selected grid cells.

Households were subsequently approached using a random walk method, with enumerators contacting every third household. In each household, one respondent aged 18 or older was selected using the most-recent-birthday method. Interviews lasted approximately 28 minutes on average. Overall, 5,288 contact attempts resulted in 2,059 completed interviews, corresponding to a response rate of 38.9%. For more details on the data, refer to Schaub et al. (2025).

4.2.1 Descriptives

For this thesis, I am using a subset of the original sample, consisting of only respondents who were at least 12 years old when the Bosnian War started in 1992 (meaning older than 43 at the time of the survey). Descriptive statistics for this sample can be found in table 1.

Name	Mean	SD	Min	Max	Count
Age	59.07	9.30	44	85	968
Male	0.53	0.50	0	1	968
Bosniak	0.70	0.46	0	1	952
Croat	0.11	0.31	0	1	952
Serb	0.19	0.39	0	1	952

Table 1: Sample descriptive statistics for key demographic indicators (age, sex, ethnicity).

Studying CRSV is challenging because it involves "thorny (…) methodological challenges in terms of how to collect data on hard-to-measure phenomena" (Nordås & Cohen, 2021, p.207). One major concern is underreporting of CRSV, primarily due to stigma and shame (Wood, 2008).

In collecting the data, we used a dual strategy to address this. Firstly, questions on sensitive items

were self administered: interviewers were asked to hand over the CAPI device, and respondents would work through these questions by themselves. The question battery respondents had to answer was introduced with the phrase: "During the period of the war in the 1990s, which of the following, if any, happened to you, to members of your family or community? I will hand over the device to you to give you privacy. If a particular event happened to you or to someone in your family or community – or you suspect that it happened – make a cross, otherwise leave empty." Below, there were several forms of victimization listed, one of which was CRSV. For every type of victimization, respondents could select "Me", "Someone in my Family", or "Someone in my community". They could also refuse to answer for any given item. Leaving an item empty was taken to mean that that item had not occurred to any of the listed people.

Second, to validate responses to sensitive items in terms of overall prevalence, we also implemented two list experiments, also known as item count technique (ICT) (Blair & Imai, 2017; Droitcour Miller, 1984).

In the first list experiment, respondents were randomly divided by the CAPI system into two equally sized groups. Those assigned to the treatment condition were asked, "Can you tell me how many of these experiences happened to you during the war in the 1990s?" and presented with a list containing four statements: "I had to leave my home", "I was part of a prisoner's exchange", "I witnessed someone being killed", and "I was subjected to sexual violence". Participants in the control condition received an otherwise identical list that excluded the statement "I was subjected to sexual violence". All respondents were instructed to indicate only the total number of listed experiences that applied to them during the 1990s conflict, and were clearly told that they did not need to specify which particular events they had experienced. The estimated prevalence of exposure to wartime sexual violence is calculated as the difference between the mean item count reported by the treatment group and that reported by the control group. The second list experiment followed the same design logic, but instead asked about the victimization of family members.

Estimates for the prevalence of CRSV in the sample from both the self-reported measure, and the list experiment, are shown in figure 11. Reults from the list experiment broadly speaking validate the self-reported measure of CRSV.

4.2.2 Conjoint Experiment

Conjoint experiments "ask respondents to chose from or rate hypothetical profiles that combine multiple attributes, enabling researchers to estimate the relative influence of each attribute value on the resulting choice rating" (Hainmueller et al., 2014, p. 2). Their basic logic is explained in section 2.4. We conducted a conjoint experiment designed to measure interpersonal trust, where respondents had to choose which person (represented by a randomized profile) they would trust more in an everyday interaction. Respondents were given two scenarios:

> **Seek-help scenario:** *Suppose you are walking down a busy street in the afternoon. Someone approaches you and asks if they could borrow your phone to make a call. I will now show you descriptions of different people. If you were in such a situation, who would you rather hand your phone to: person A or person B?*
>
> **Offer-help scenario:** *Suppose you are traveling on a train in the afternoon and you*

Estimand	Victimization	Estimate	2.5%	97.5%
List experiment	Self	2.40	-1.57	6.37
List experiment	Family	9.57	6.25	12.90
Self-reported	Self	2.38	1.58	3.55
Self-reported	Family	12.50	10.60	14.70

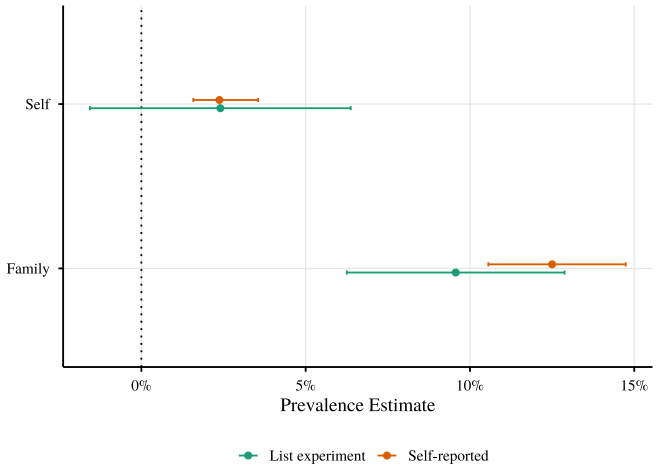


Figure 11: Prevalence of CRSV, self-reported and as estimates from list experiment. Confidence intervals for the self-reported measure were constructed on the logit scale.

need to use the restroom. You have heavy luggage with you that is difficult to carry. I will now show you descriptions of different people. If you were in such a situation, who would you rather ask to look after your belongings: person A or person B?

For each scenario, respondents had to complete six choice tasks. We always presented them with two randomly generated profiles, and asked them to choose which person they would prefer to trust in the given situation. An example of such a choice task is shown in figure 12. The attributes and values used to randomly generate profiles are shown in table 2.

Attribute	**Values**		
Age	23	49	67
Class	Lower	Middle	Upper
Education	None	Highschool	University
Ethnicity	Bosniak	Croat	Serb
Sex	Female	Male	

Table 2: Attributes and values used in the conjoint experiment.

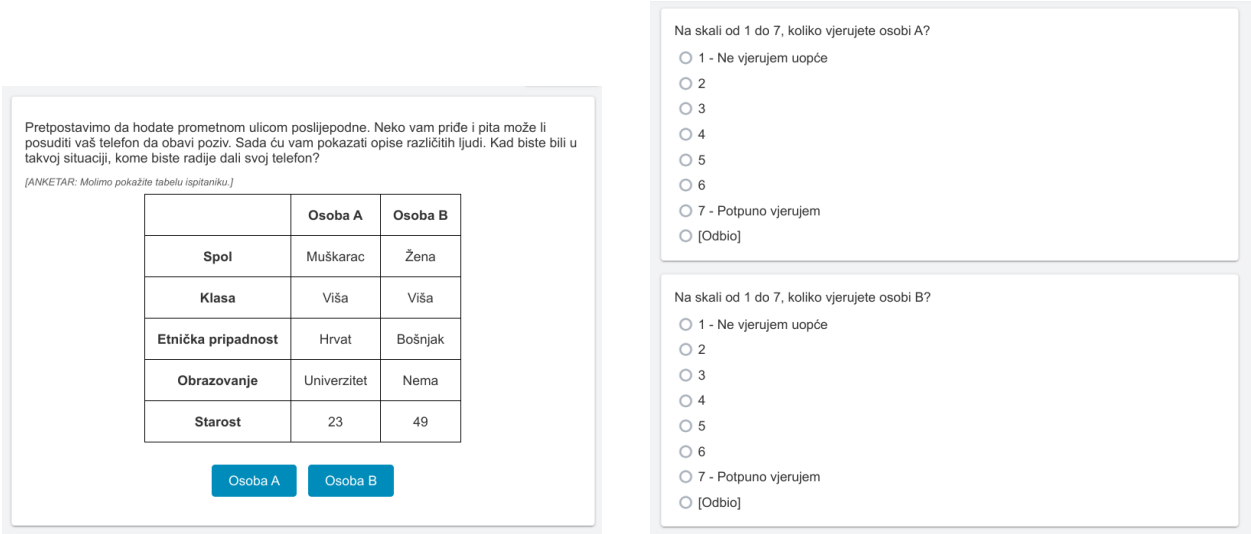


Figure 12: An example conjoint task from the survey: on the left the choice task, and on the right the rating task.

AMCEs and Marginal Means for all attributes are shown in figure 13. There are a number of interesting findings here which are only of secondary interest to this case study, such as males, less educated people, and young people being less trusted, while more educated people and older people are being trusted more. Of interest here is to establish the basic impact of ethnic group membership on interpersonal trust: profiles belonging to a different group than the respondent are significantly less likely to be selected in the first place, while the profiles belonging to the ingroup are more likely to be trusted: there is a clear ingroup premium visible.

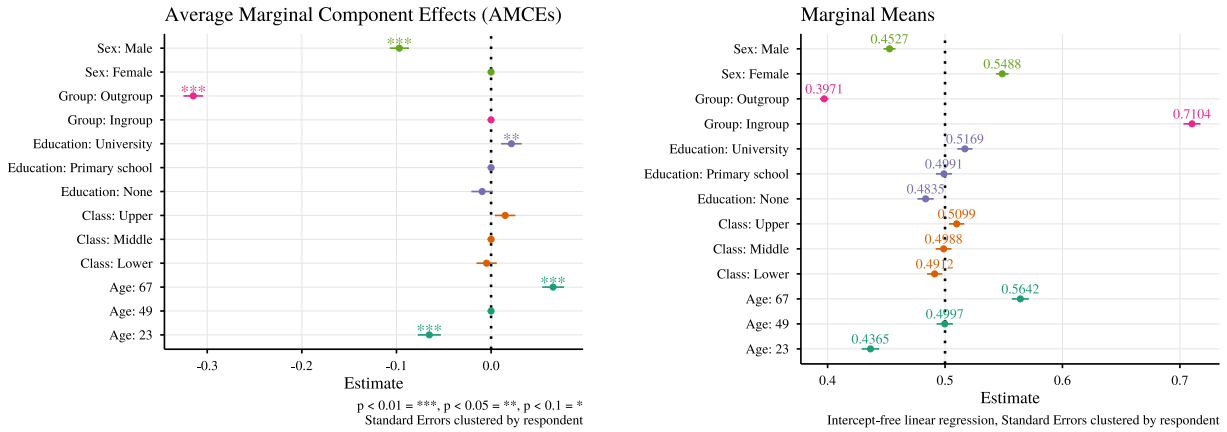


Figure 13: AMCEs and Marginal Means from the conjoint experiment.

4.3 Methodology

The outcome of interest is whether a given profile was selected in a choice task, meaning the hypothetical person described in that profile was trusted by the respondent. The main treatment is ethnic

group membership, meaning whether a profile belongs to the same ethnic group as the respondent, or a different ethnic group. This treatment variable was created by dichotomizing the "Ethnicity"-attribute from the profiles: it was set to 1 if the profile belonged to the same ethnic group as the respondent, and 0 if it did not. The expectation is that, due to randomization, all other attributes vary randomly across ingroup and outgroup profiles, and the ATE for ingroup membership is thus identified as

$$\mathbb{E}[\text{Selected} \mid \text{Ingroup}] - \mathbb{E}[\text{Selected} \mid \text{Outgroup}].$$

As can be seen, both of these separate expectations correspond to a marginal mean (Leeper et al., 2020), and their pairwise difference thus has a direct causal interpretation as an AMCE (Hainmueller et al., 2014). The effect modifier of interest is exposure to CRSV. For the context of this experiment, I define exposure to CRSV as either the respondent or a family member of the respondent being subjected to CRSV, as per the respondent. As such, the measure defines exposure in a broader sense, including both direct and indirect exposure.[15] CRSV is non-random, and due to its nature not physically randomizable, and thus selection into CRSV is subject to several intervening, or confounding variables, that plausibly also affect the main treatment effect.

First, and most obviously, the sex of the respondent plays a crucial role: women are much more likely to become the victims of CRSV (Nordås & Cohen, 2021), and men and women also exhibit differences in intergroup trust, and trust patterns in general (Maddux & Brewer, 2005). Second, victimization differed markedly across ethnic groups (Schaub et al., 2025), and different ethnic groups might also trust each other differently. Next, the socio-economic background of a respondent might also play a role in both victimization, as the cost of conflicts is likely unevenly distributed across socio-economic backgrounds, and people from different socio-economic backgrounds might trust differently. I use respondents' education level as a proxy, given that for most older respondents, its somewhat plausibly pre-victimization, and thus a marker of prior socio-economic status (SES). Other markers of SES are likely to be affected by CRSV itself, and might be part of the effect. Lastly, exposure to other forms of victimization than CRSV during the conflict might play a role for both exposure to CRSV, and for intergroup trust. Summarizing these background factors as $\mathbf{X}$, the quantity of interest is now

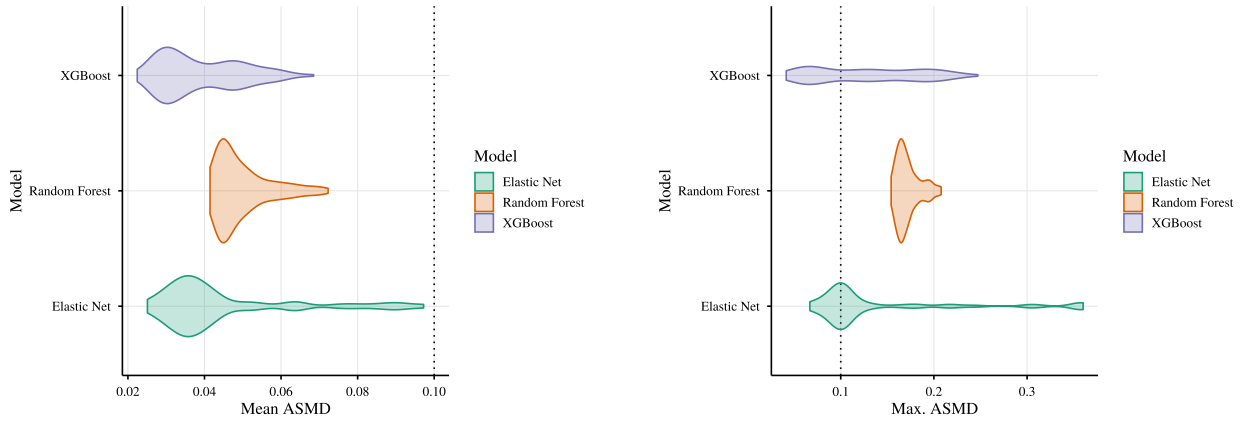
$$\begin{aligned} &\mathbb{E}_{\mathbf{X}}\Big[\mathbb{E}[\text{Selected} \mid \text{Ingroup, CRSV}, \mathbf{X}] - \mathbb{E}[\text{Selected} \mid \text{Outgroup, CRSV}, \mathbf{X}]\Big] \\ -&\mathbb{E}_{\mathbf{X}}\Big[\mathbb{E}[\text{Selected} \mid \text{Ingroup, No CRSV}, \mathbf{X}] - \mathbb{E}[\text{Selected} \mid \text{Outgroup, No CRSV}, \mathbf{X}]\Big] \end{aligned} \tag{10}$$

which is an analogue to eq. 7.

[15] While this risks conflating the effects of direct and indirect exposure, the prevalence of direct exposure is very low, and since the primary purpose of this case study is to show the implementation of the proposed method, I decided on a joint measure.

4.3.1 Propensity score estimation

I considered three different models for estimating the propensity score for S, or π_S: Elastic Net regression, Random Forest, and XGBoost (Chen & Guestrin, 2016). I tuned all models directly on the achieved covariate balance, as described in algorithm 1. For each model configuration, I computed mean and maximum absolute standardized mean differences (ASMDs), as well as effective sample size (ESS) for $S = 1$ and $S = 0$ units. The distribution of mean and maximum ASMDs by model across all configurations can be found in figure 14. More visualizations are shown in the appendix. Results show that for each model, there exists some hyperparameter configuration that yields a maximum ASMD below 0.1, but for the Random Forest there exists no configuration with a maximum ASMD below 0.1. All models manage to achieve a mean ASMD below 0.1.



(a) Distribution of mean ASMDs by model.

(b) Distribution of maximum ASMDs by model.

Figure 14: Distribution of maximum ASMDs by model.

I then fit the best configuration of each model to the data & calculated the final imbalance, which can be found in table 3. I also included a plain logistic regression as a benchmark, as researchers still most commonly rely on logistic regression to estimate propensity scores (cf. Leite et al., 2025).

Model	Mean ASMD	Max ASMD	ESS (S=1)	ESS (S=0)
XGBoost	0.0224	0.0473	51.0	805
Elastic Net	0.0251	0.0669	65.8	827
Random Forest	0.0414	0.1600	85.7	828
Logistic Regression	0.0448	0.1020	42.4	814

Table 3: Imbalance and ESS by model, sorted by mean ASMD.

After fitting the best models to the data, I also considered the imbalance reduction achieved by each model for each covariate separately, which is shown in figure 15. Generally, adequate balance is considered to be achieved when standardized mean differences are below 0.1 (Austin, 2009, 2019; Cohen, 1988; Mamdani et al., 2005); the vertical lines indicate this threshold.

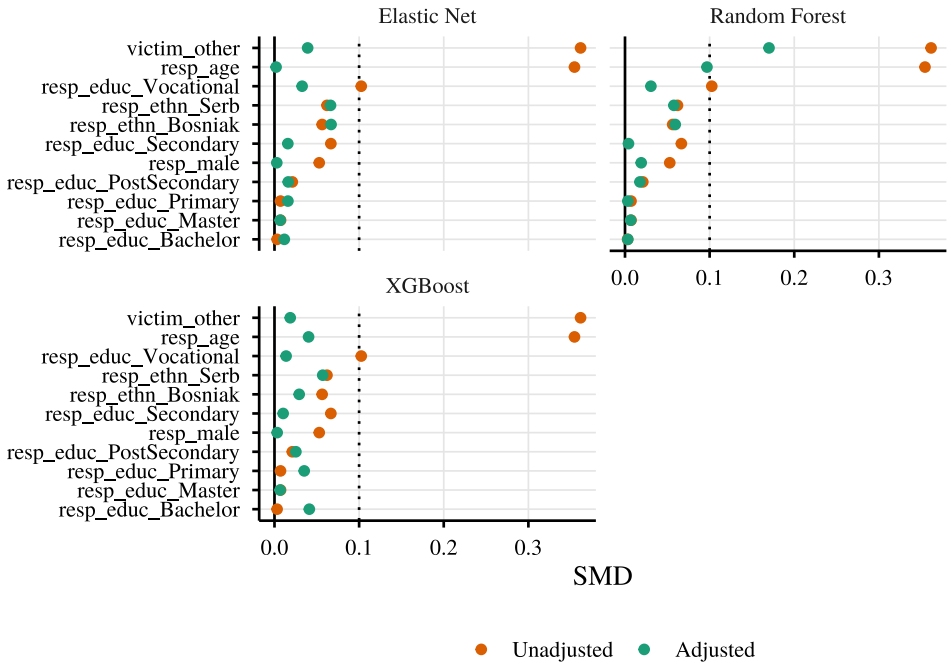


Figure 15: Imbalance reduction for all covariates by model.

Overall, XGBoost provided the best-balancing propensity score estimates, so I decided to use them for the analysis. All evaluated methods outperformed a plain logistic regression.

4.3.2 Models

I first fit a naive model

$$\text{Selected}_{i,j} = \beta_0 + \beta_1 \text{Ingroup}_{i,j} + \beta_2 \text{CRSV}_i + \beta_3 (\text{Ingroup}_{i,j} \times \text{CRSV}_i) + \varepsilon_i$$

where β_3 provides an estimate of effect modification by CRSV. However, as established earlier, β_3 has no causal interpretation, it denotes simply the observed difference. Next, I fit a regression model properly controlling for potential confounders

$$\text{Selected}_{i,j} = \beta_0 + \beta_1 \text{Ingroup}_{i,j} + \beta_2 \text{CRSV}_i + \beta_3 (\text{Ingroup}_{i,j} \times \text{CRSV}_i) + \mathbf{X}_{i,j}^{\top} \boldsymbol{\gamma}_1 + \text{Ingroup} \cdot \mathbf{X}_{i,j}^{\top} \boldsymbol{\gamma}_2 + \varepsilon_i$$

where $\mathbf{X}$ includes the respondent i's ethnicity, education level, sex, and whether they have experienced victimization other than SV during the war. I then fit the same model on data that was weighted according to the pooled IPW scheme outlined in this thesis. I used propensity score estimates from an XGBoost model, which was able to attain the best covariate balance between the exposed and non-exposed group. The reason for fitting the model including controls to the data despite it being weighted is that, despite overall very good balance achieved, we know there to

be residual imbalance between the two groups, and we thus know the simple difference-in-means estimator to be biased.

Lastly, I also conducted propensity score matching, matching exposed units to non-exposed using the propensity score estimates from XGBoost. The matching estimate is not directly comparable to the other models, as the targeted estimand is the ATT, but serves as additional test for causal effect modification. A balance plot for the matching is shown in the appendix, in figure 28.

For all analyses, I pool data from both scenarios. F-tests confirm that neither the overall pattern of attribute effects ($F = 1.1065$, $p = 0.3551$), nor the ingroup effect specifically ($F = 0.0764$, $p = 0.9265$) differs systematically across scenarios, justifying this decision.

4.4 Results & Discussion

Estimates from all models are shown in table 4. The naive model provides a baseline estimate of effect modification by CRSV, but as established, the interaction coefficient carries no causal interpretation since it does not account for the confounders driving selection into CRSV exposure. Moving to the controlled regression, which adjusts linearly for confounders in both the outcome and the treatment effect heterogeneity, already produces meaningful attenuation, shrinking the coefficient. Under the identifying assumptions outlined above, this remaining difference represents the effect modification plausibly attributable to CRSV itself, net of linear confounding by the observed covariates.

The pooled IPW estimate attenuates the coefficient further, suggesting that the controlled regression does not fully account for the confounding structure — consistent with the presence of mild nonlinearity in how the covariates drive selection into CRSV exposure, which the parametric model is not equipped to capture but the flexible propensity score model can. That the IPW estimate remains statistically significant despite this additional adjustment lends credibility to the conclusion that the effect modification is not purely an artifact of confounding. The matching estimator, which targets a conditional ATT and is therefore not directly comparable to the other models, similarly returns a significant and substantively large interaction effect, providing further corroboration.

Taken together, the interaction coefficient on Ingroup × CRSV is negative and significant across all four specifications, which we can interpret as exposure to CRSV causally reducing the degree to which respondents favor ingroup members in trust decisions. Inspecting the group-level estimates plotted in figure 16 helps clarify the mechanism underlying this pattern: the difference is driven primarily by a decline in ingroup trust among those exposed to CRSV, rather than by any systematic change in outgroup trust. Indeed, differences in outgroup trust across exposure groups largely disappear once confounding is accounted for, suggesting that CRSV mostly erodes the ingroup trust premium.

	Naive	Controlled reg.	IPW	Matched
Ingroup	0.327***	0.116	0.092	0.334**
	(0.010)	(0.076)	(0.075)	(0.157)
CRSV	0.033*	0.028	0.009	0.018
	(0.017)	(0.018)	(0.018)	(0.024)
Ingroup × CRSV	-0.111***	-0.099***	-0.067**	-0.105**
	(0.032)	(0.030)	(0.030)	(0.042)
Num. Obs.	11290	11290	11290	2532
AIC	37923.3	37855.1	38576.3	8505.0
BIC	120668.6	120439.1	121160.3	23143.6

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Estimates of effect modification from the different models. Standard errors are clustered by respondent. The coefficient on Ingroup is not causally meaningful anymore when identification targets the interaction term, it should only be given causal interpretation in the naive model. The matching estimator delivers a conditional ATT estimate.

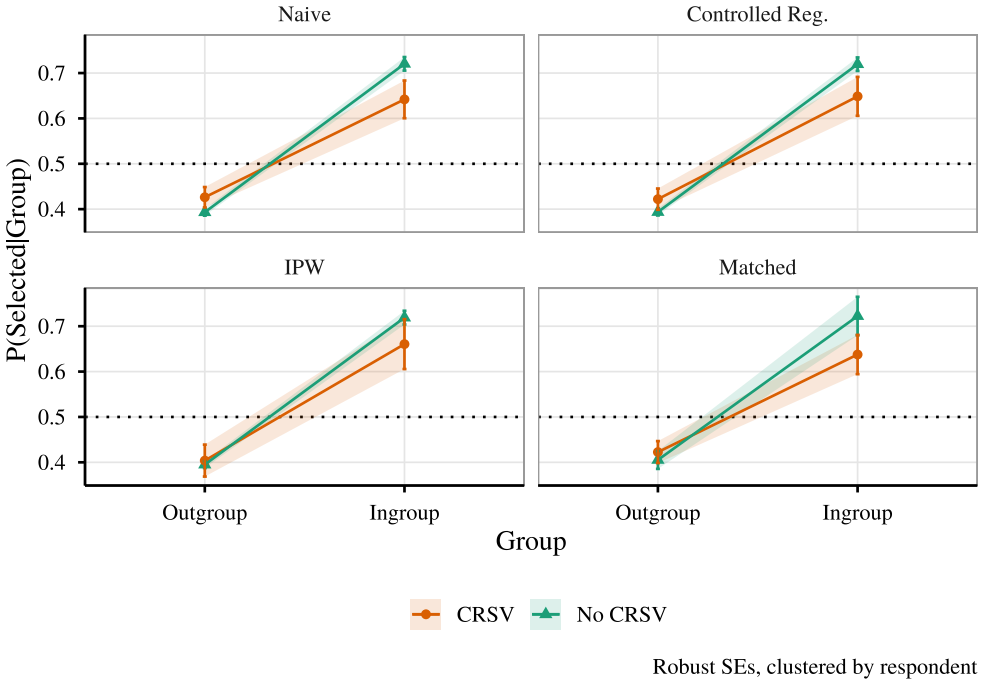


Figure 16: Estimates for ingroup and outgroup trust, by exposure to CRSV.

5 Concluding Remarks

This thesis set out to address a problem that frequently arises in applied social science research: how to estimate causal treatment effect modification when the effect modifier of interest was not randomized, and observed differences in treatment effects may therefore reflect covariate imbalance rather than genuine causal interaction. I gave special attention to the conjoint experiment, which is a commonly used experimental design in experimental political science research The proposed solution uses inverse probability weighting as nonparametric preprocessing, combining a flexible propensity score model, tuned directly for covariate balance, with a simple parametric outcome model estimated on the weighted data. The approach retains the interpretability of standard regression-based analyses while reducing dependence on restrictive functional form assumptions about how covariates drive selection into the effect modifier or vary the treatment effect.

The case study illustrated the method's implementation using original data from a conjoint experiment conducted in Bosnia and Herzegovina, examining whether exposure to conflict-related sexual violence (CRSV) causally modifies the effect of ethnic group membership on interpersonal trust. Across all model specifications, exposure to CRSV was associated with a significant reduction in the ingroup trust premium, with the IPW estimate showing somewhat more attenuation than the controlled regression, suggesting the presence of mild nonlinearity in the confounding structure. The consistency of the result across naive, controlled, weighted, and matched estimators lends credibility to the finding that CRSV causally varies intergroup trust, rather than merely being a correlate of characteristics that do.

Taken together, the results suggest that the proposed method is a useful and practical tool for researchers who wish to move beyond describing treatment effect heterogeneity and toward making causal claims about its sources, without sacrificing interpretability or requiring fully nonparametric outcome modeling.

Future work seems most promising with regard to propensity score estimation. It should explore directly optimizing model parameters for balance, rather than merely tuning hyperparameters, which may yield significant improvements in the quality of the resulting propensity score estimates, and thus the reliability of estimates based on these propensity scores. The use of flexible models to estimate propensity scores also needs to be evaluated more generally. Currently, there are plenty of conventional wisdoms, but little empirical evidence to back them up. Further, awareness needs to be raised that the "best" propensity score model is highly data specific, and what works very well in one context might deliver subpar results in another.

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A Appendix: Graphical Perspectives on Effect Dependency

The problem of how to show effect dependency in DAGs has been given considerable thought in the literature. The simplest way of displaying effect dependency in a DAG is to treat them as agnostic towards effect dependency, as e.g. VanderWeele and Robins (2007) do: S is considered a (direct) effect modifier of the causal effect of T on Y because it is a direct cause of Y (VanderWeele & Robins, 2007, p. 563). Such a "simple" DAG is shown in figure 17.

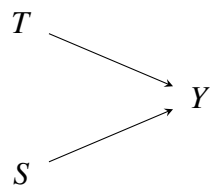


Figure 17: A normal DAG, which is agnostic to effect modification.

However, as Weinberg (2007) notes, in this DAG there is "no graphical representation to indicate that a factor is regarded as an effect modifier for some other (...) factor" (p. 2). They instead suggest drawing "edges into edges", as shown in figure 18. But a fundamental drawback of this approach is that this is not a valid DAG, and in fact not even a valid graph, as edges should only connect nodes, not edges (recall that in a DAG, $\mathbf{E}$ is the set of directed edges $X \to Y$ where $X, Y \in \mathbf{V}$, the set of nodes). As such, it is incompatible with known algorithms e.g. for deriving conditioning sets: if we introduced more variables it would not be obvious how to causally identify for example the second-order effect $S \to (T \to Y)$ from standard DAG theory.

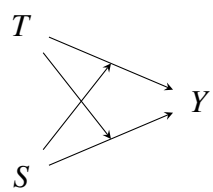


Figure 18: A DAG modified to show effect modification by drawing edges into edges, invalidating the DAG.

Another alternative was suggested by Nilsson et al. (2021), who devised the "IDAG", a separate graph that should be displayed next to the DAG, where edges - or effects - become nodes themselves, allowing arrows into them, enabling a graphical representation of "effects on effects". An example of a DAG and an associated IDAG derived from Nilsson et al. (2021) is shown in 19. While easy to understand intuitively, if S modifies the effect of T on Y, that also implies that T modifies the effect of S on Y, since interaction is symmetric (cf. Weinberg, 2007), which is absent from this representation. Every edge from the original DAG would need its own IDAG, which quickly grows impractical. However, one merit of this approach is that it shows very clearly that we are dealing with a second-order estimand, an effect on an effect, so to speak.

(a) DAG *(b) IDAG*

Figure 19: DAG and one corresponding IDAG.

Lastly, one could add an "interaction node" to the DAG, which shows the joint effect of two variables that is distinct from their separate effect, which is an idea from Attia et al. (2022). Here, we introduce $T \times S$, the separate joint effect of T and S, as a descendant of both variables, while both also retain separate effects that are not subject to modification from the other variable. An example can be found in figure 20. However, it might quickly clutter DAGs, as the total number of nodes scales with $\binom{n}{2}$, where n is the number of nodes representing actual variables - that is if we only consider first-degree interactions.

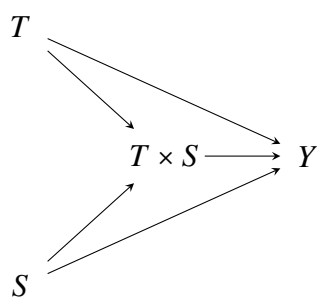


Figure 20: Interaction node representing the joint effect of T and S.

I follow VanderWeele and Robins (2007) in sticking with standard DAGs, without extra edges, graphs, or nodes attached, as DAGs make no claims about the functional form of the relationships between variables.

B Appendix: Simulation Study

Having defined the estimand and outlined the estimation strategy, I now evaluate the proposed approach on simulated data before turning to the empirical application. The simulation study pursues three objectives. First, I assess whether the proposed IPW estimator reliably accounts for confounding by $\mathbf{X}$, including nonlinear confounding, given an approximately correctly specified propensity score model. Second, I examine whether it can correctly distinguish surrogate effect modification from confounded causal effect modification: that is, whether it returns estimates near zero when S is merely a proxy for $\mathbf{X}$, and recovers the true effect when genuine causal modification is present

but confounded. Third, I compare pooled and parallel IPW estimation (Bansak, 2021) to assess whether the additional complexity of the parallel approach offers any practical benefit.

B.1 Setup

I simulated a scenario with a binary treatment $T_i \in \{0, 1\}$, a binary effect modifier S_i, and one continuous covariate X_i. The treatment was randomized independently of all other covariates:

$$T_i \sim \text{Bernoulli}(0.5)$$

The effect modifier S_i is not randomized, and is generated as a scenario-specific function $f(\cdot)$ of X_i:

$$\begin{aligned} S_i \mid X_i &\sim \text{Bernoulli}(\text{logit}^{-1}(f(X_i))) \\ X_i &\sim \mathcal{N}(0, 1) \end{aligned}$$

Potential outcomes satisfy

$$\begin{aligned} Y_i(0) &\sim g(X_i) + \varepsilon_i,\ \varepsilon \sim \mathcal{N}(0, 0.5^2), \\ Y_i(1) &= Y_i(0) + h(X_i) + \theta S \\ Y_i(1) - Y_i(0) &= h(X_i) + \theta S \end{aligned}$$

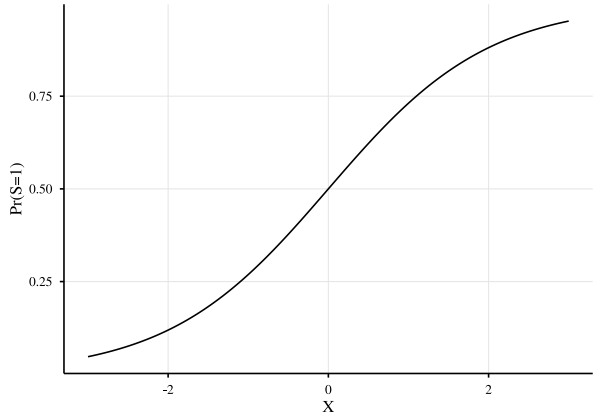
So that the treatment effect is a function of X and S. θ is the strength of the true causal effect modification by S. The realized outcome is

$$Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0).$$

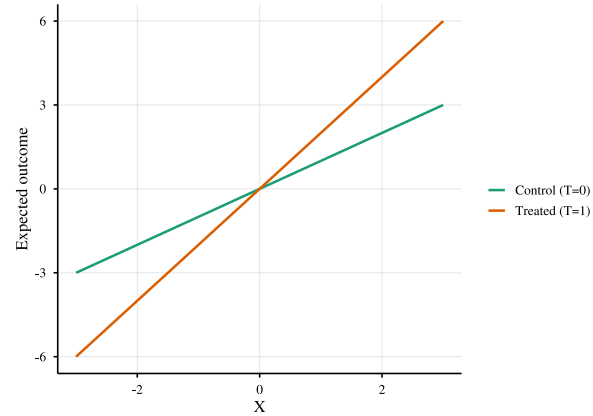
I consider two settings, $\theta = 0$ (no causal effect modification by S; it is a surrogate effect modifier) and $\theta = 0.5$ (causal effect modification by S with confounding by X). First, for each setting, I test a linear scenario

$$f(x) = x,\ g(x) = x,\ h(x) = x$$

where selection into $S = 1$ is caused linearly by X, and the treatment effect varies as a linear function of X. Visualizations can be found in figure 21.



(a) Selection into S as a function of X in the linear scenario.



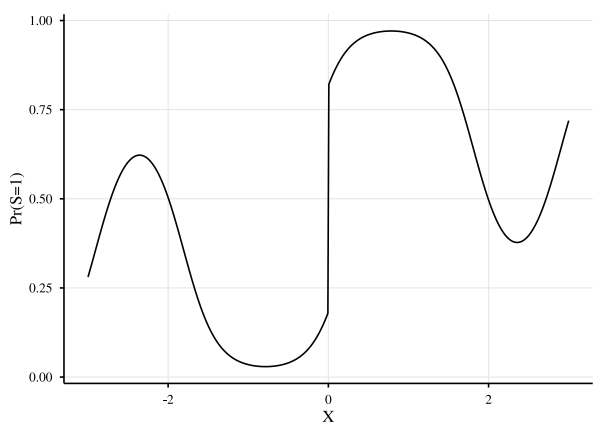
(b) Expected outcomes of treatment and control group as a function of X in the linear scenario.

Figure 21: The linear scenario.

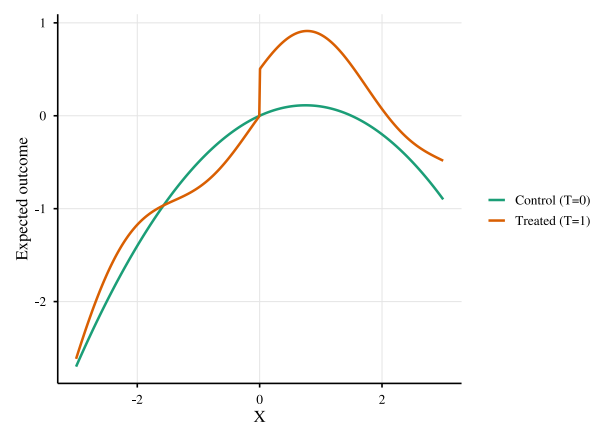
Second, I also test a nonlinear scenario

$$f(x) = \begin{cases} 3 + 2\sin(2x) - 1.5 \text{ if } x > 0 \\ 2\sin(2x) - 1.5 \text{ otherwise.} \end{cases}, \; g(x) = 0.3x - 0.2x^2, \; h(x) = \begin{cases} 0.5 + 0.3\sin(2x) \text{ if } x > 0 \\ 0.3\sin(2x) \text{ otherwise.} \end{cases}$$

where selection into S and the treatment effect are affected by X non-linearly. As can be seen in figure 22, this causes a setting where the treatment is only really effective in a specific region of X.



(a) Selection into S as a function of X in the nonlinear scenario.



(b) Expected outcomes of treatment and control group as a function of X in the nonlinear scenario.

Figure 22: The nonlinear scenario.

B.2 Models

For each scenario, I tested four models. Two linear models, one deliberately misspecified, and two IPW estimators, one "pooled", as suggested above, and one "parallel", as suggested by Bansak (2021).

B.2.1 Linear models

Starting with the simplest model, I specified a linear model that does not model effect heterogeneity caused by X instead of S explicitly (includes no interaction term $T \times X$), similar to Dong (2015) and Lee and Lee (2025), to show that such a model misattributes variation in treatment effects caused by X to S instead, if X and S are correlated and $T \times X$ not included in the model, as

$$Y = \beta_0 + \beta_1 T + \beta_2 S + \beta_3 (T \times S) + \beta_4 X$$

where $\beta_3 (T \times S) = \hat{\theta}_{\text{Linear, misspec.}}$. The second, correctly specified linear model, includes that extra interaction term:

$$Y = \beta_0 + \beta_1 T + \beta_2 S + \beta_3 (T \times S) + \beta_4 X + \beta_5 (T \times X)$$

where again $\beta_3 = \hat{\theta}_{\text{Linear}}$. As long as relationships are approximately linear, this model should deliver a consistent estimate of causal effect modification by S, since it correctly attributes variation in treatment effects caused by X to X via $\beta_5 T \times X$.

B.2.2 IPW estimators

For both IPW estimators, the propensity score was modelled using natural cubic splines with a logistic regression. The flexibility of the splines is determined by the degrees of freedom `df`; no "right" value for this parameter can be analytically determined from the data, so it is a hyperparameter which has to be set prior to - and remains constant during - training. As such, the degrees of freedom were tuned for each model using the tuning procedure layed out above to determine the value (i.e. the amount of flexibility allowed) that would produce the best balance. Based on the propensity scores of the best model found in tuning, I then calculated stabilized IPWs w_i, which were used in a weighted least squares regression for the outcome

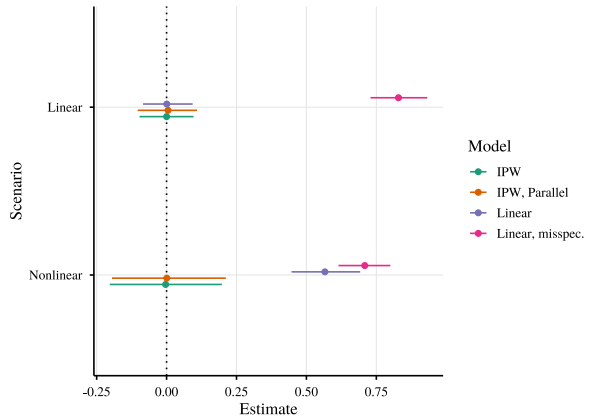
$$\underset{\beta_0, \beta_1, \beta_2, \beta_3}{\arg\min} \sum_i w_i (Y_i - \beta_0 + \beta_1 T + \beta_2 S + \beta_3 (T \times S) + \beta_4 X + \beta_5 (T \times X))^2$$

which essentially means fitting the same parametric model a researcher would normally be inclined to estimate (the correctly specified linear regression) on the weighted data. Here, $\beta_3 = \hat{\theta}_{\text{IPW}}$. For the parallel estimator (Bansak, 2021) I first split the data by treatment assignment; I then tuned models, calculated weights & fit outcome models regressing Y on S separately for $T = 1$ and $T = 0$ units. The difference in the coefficients on S between treated and untreated units is then taken as

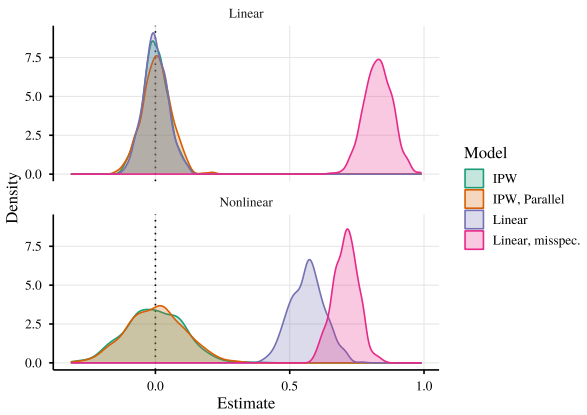
estimate of causal effect modification by S.

B.3 Results

B.3.1 First setting: surrogate effect modification



(a) Mean estimates for each model in each scenario, with empirical 95% confidence intervals.



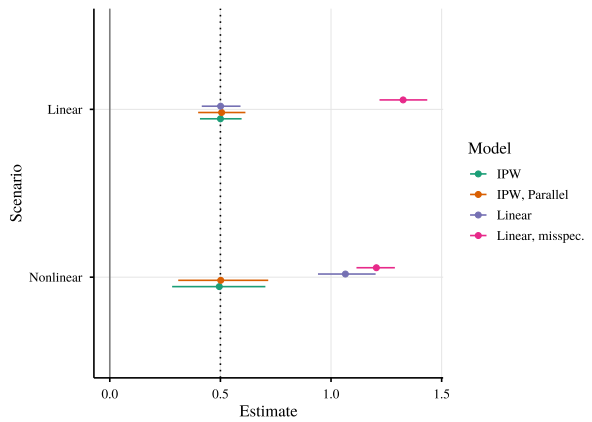
(b) Distributions of the estimates by model and scenario, for the first setting where S does not causally modify the effect of T on Y (it is only a surrogate effect modifier and all heterogeneity is due to X).

Figure 23: Visualizations of the simulation results.

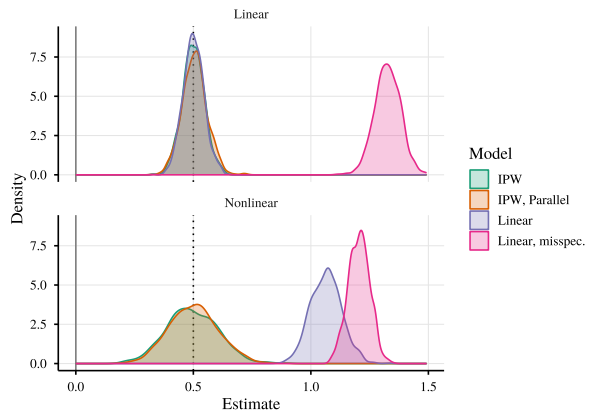
Results of the first setting, $\theta = 0$, are shown in figure 23. As can be seen, even under complete linearity, the linear model that does not model an interaction between T and X severely overestimates the true effect modification of $T \rightarrow Y$ by S, as it falsely attributes the variation in the treatment effect caused by X to S. Differences between the other three models are negligable. Under nonlinearity of selection into S and the treatment effect heterogeneity caused by X, the correctly specified linear model also collapses. The pooled and parallel IPW estimators both remain consistent, as long as the propensity score model is correctly specified. This highlights the strengths of IPW in this setting: it allows combining a potentially very flexible model for the propensity score with a simple parametric outcome model familiar to applied researchers. In general, if the linearity assumption holds for both selection into S and X's effect on the treatment effect, the correctly specified linear model is the most efficient estimator. If linearity does not hold, the suggested IPW-based estimator outperforms. Since pooled IPW is simpler to estimate and allows for easy uncertainty estimation (we are left with a full outcome model, not just a point estimate), pooled IPW might be preferrable to parallel, especially as the procedure for modeling the propensity score becomes more complex. All in all, the first setting shows that IPW estimators can reliably detect surrogate effect modification.

B.3.2 Second setting: confounded causal effect modification

Figure 24 shows the results for the second setting, $\theta = 0.5$, where there is a true causal effect modification by S, but it is confounded by X. Results are overall identical to the first setting. The idea in this setting with confounded (causal) effect modification is to assess whether estimators systematically over- or underestimate causal effect modification by S under confounding. For example, a model that yields estimates of the true causal effect modification biased towards zero would still seem to perform properly in the first setting.



(a) Estimates with empirical 95% confidence intervals. Solid line indicates zero, dashed line indicates true value again.



(b) Distributions of the estimates by model and scenario.

Figure 24: Visualizations of the simulation results for the second setting, where S does causally modifiy the effect of T on Y, but the relationship is confounded.

Results show that this is not the case: IPW estimators recover the true effect, even under complex nonlinear confounding given an approximately correct propensity score model. To sum up, the IPW estimators (1) properly identify surrogate effect modification (no true effect modification), and (2) deliver consistent estimates of true causal effect modification when the modification is confounded. Lastly, in the setting evaluated, parallel estimation does not offer any advantage over pooled estimation of the propensity score, and pooled estimation might thus be preferrable due to reduced overhead and its more straightforward estimation procedure.

C Appendix: Propensity Score Estimation

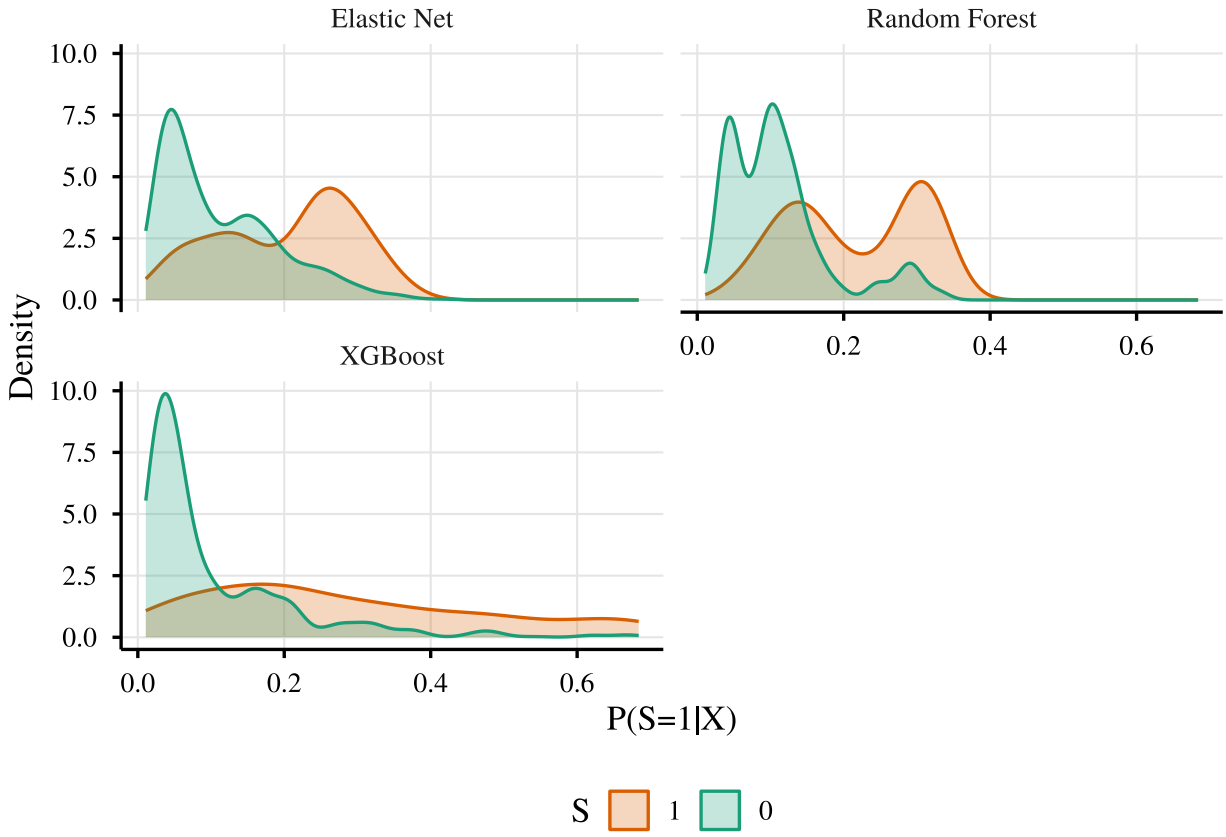


Figure 25: Overlap in the distributions of the groups of S for each propensity score model evaluated.

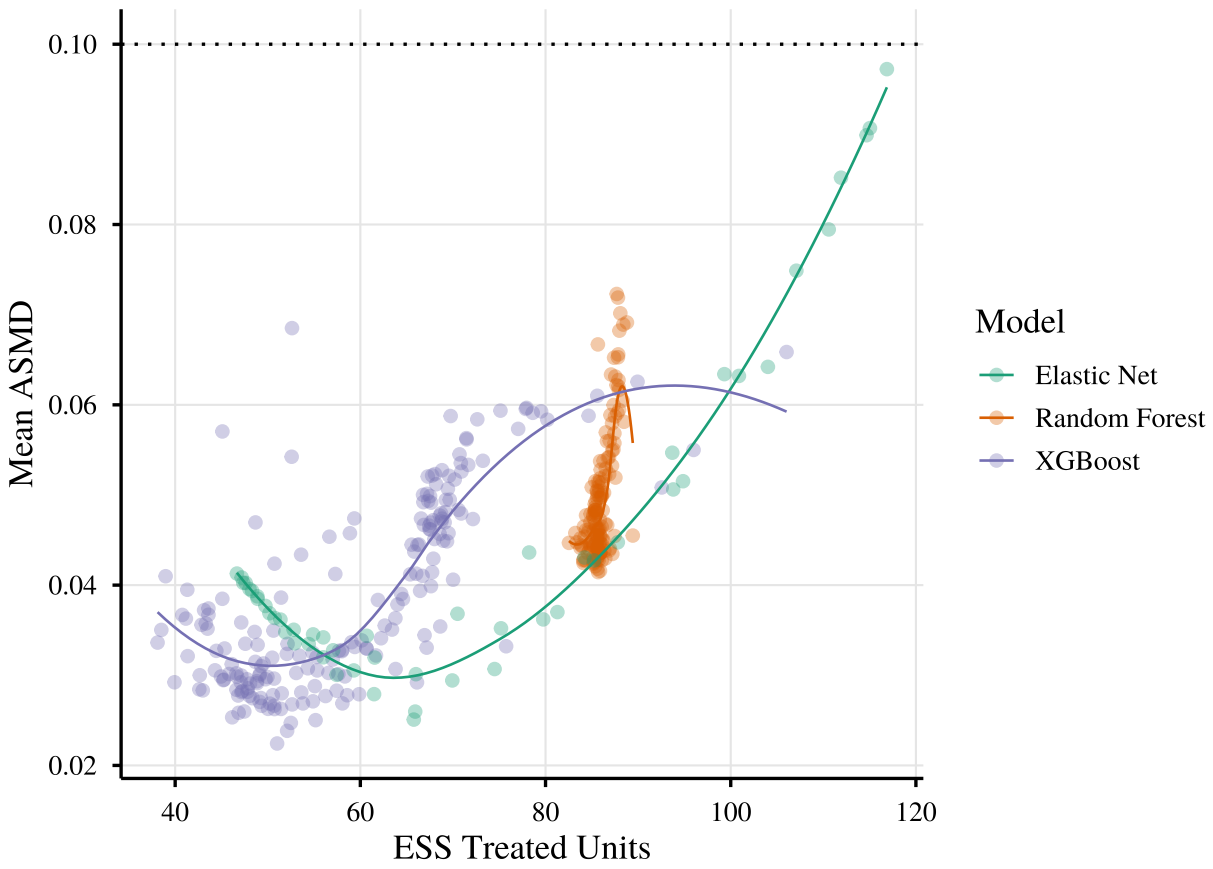


Figure 26: Effective sample size vs. mean ASMD. Here, we can check whether models that achieve better balance do so at the cost of effective sample size, effectively dropping many observations in the process.

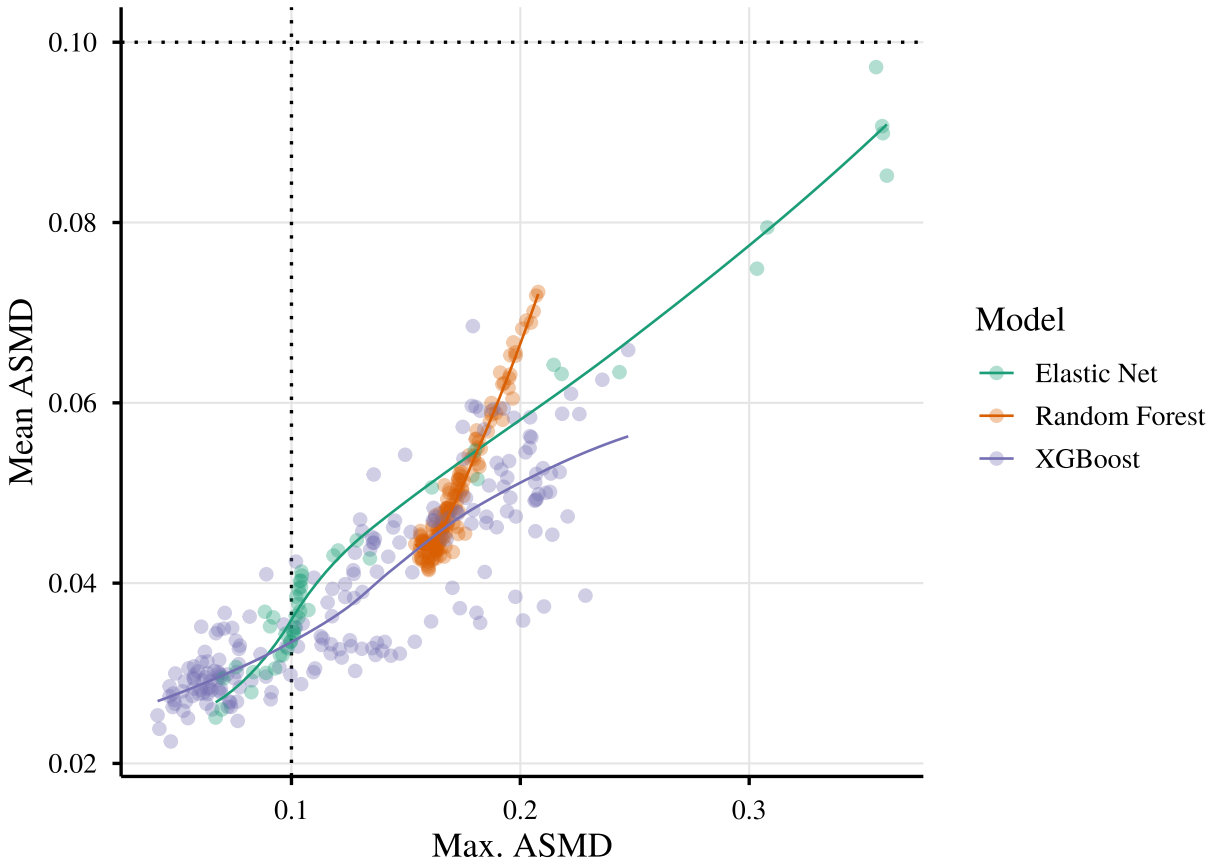


Figure 27: This visualization shows mean ASMD contrasted against maximum ASMD. We can use it to assess whether models exhibit "sacrifice one"-behavior, where they "sacrifice" some coviarate to achieve better mean ASMD.

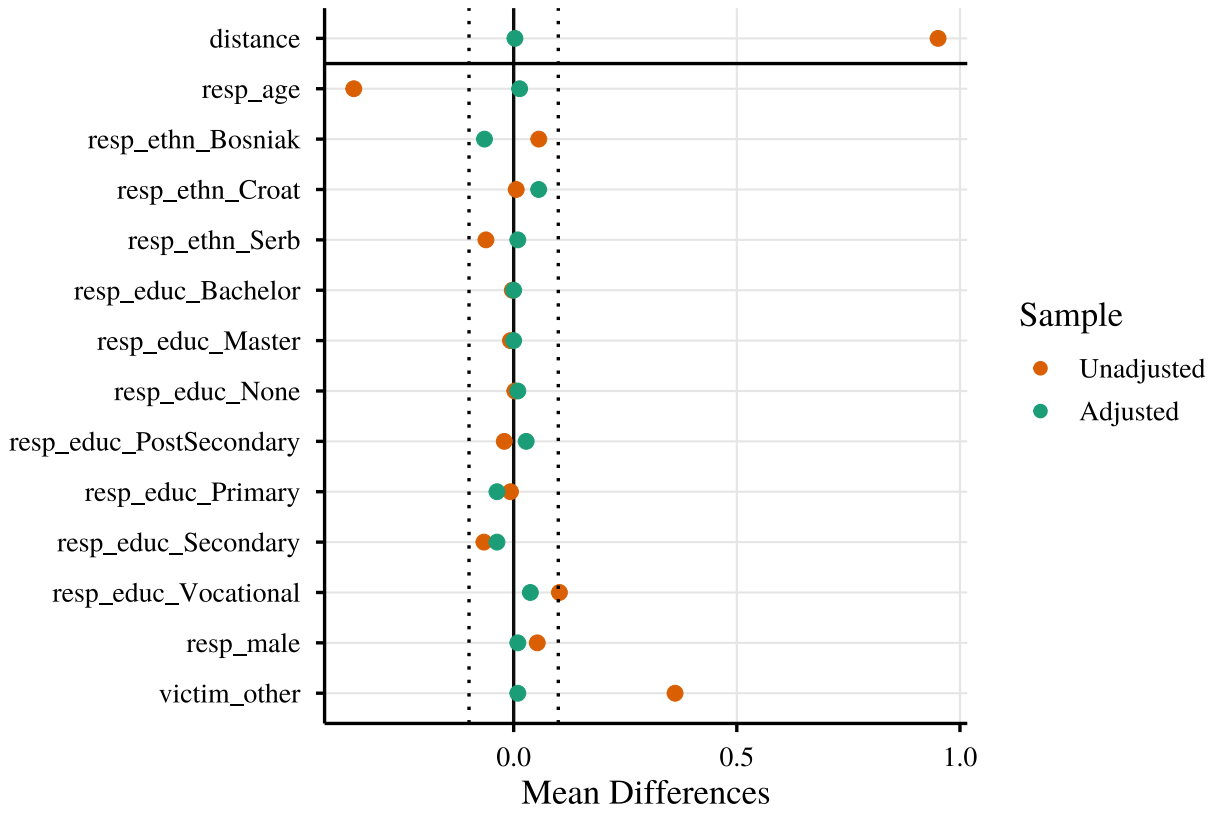


Figure 28: Covariate balance before and after propensity score matching.